

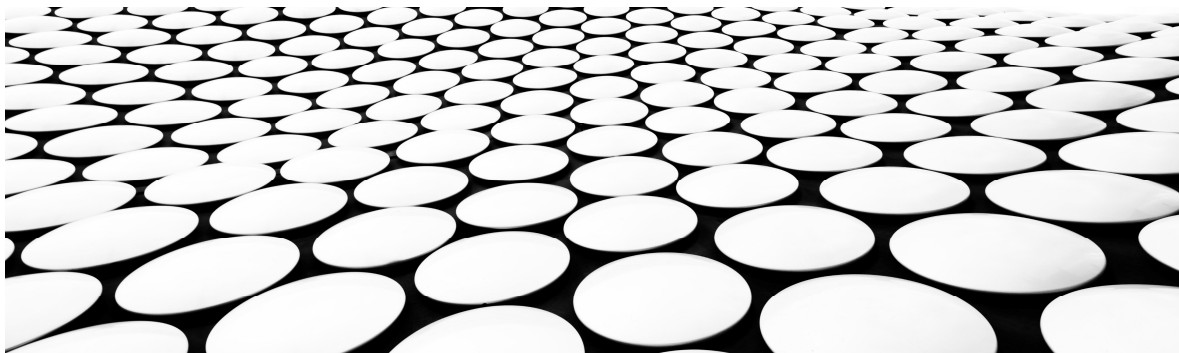


ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN
KHOA ĐIỆN TỬ - VIỄN THÔNG

THIẾT KẾ VI MẠCH ĐIỆN TỬ

CHAPTER 3: MOSFET MODELS

LÊ ĐỨC HÙNG



CONTENT

- Introduction to Modeling
- Transistor MOSFET Models
- Capacitor Models
- Noise Model

INTRODUCTION TO MODELING

FET Model Generation

- First Generation – Physically based analytical model including all geometry dependence.
- Second Generation – Model equations became subject to mathematical conditioning for circuit simulation. Use of empirical relationships and parameter extraction.
- Third Generation – A return to simpler model structure with reduced number of parameters which are physically based rather than empirical. Uses better methods of mathematical conditioning for simulation including more specialized smoothing functions.
- Performance Comparison of Models (from Cheng and Hu, MOSFET Modeling & BSIM3 Users Guide)

Model	Minimum L (μm)	Minimum Tox (nm)	Model Continuity	i_D Accuracy in Strong Inversion	i_D Accuracy in Subthreshold	Small signal parameter	Scalability
MOS1	5	50	Poor	Poor	Not Modeled	Poor	Poor
MOS2	2	25	Poor	Poor	Poor	Poor	Fair
MOS3	1	20	Poor	Fair	Poor	Poor	Poor
BSIM1	0.8	15	Fair	Good	Fair	Poor	Fair
BSIM2	0.35	7.5	Fair	Good	Good	Fair	Fair
BSIM3v2	0.25	5	Fair	Good	Good	Good	Good
BSIM3v3	0.15	4	Good	Good	Good	Good	Good

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INTRODUCTION TO MODELING

First Generation Model

- Level 1 (MOS1)
 - Basic square law model based on the gradual channel approximation and the square law for saturated drain current.
 - Good for hand analysis and hand calculations.
 - Needs improvement for deep-submicron technology (must incorporate the square law to linear shift)
- Level 2 (MOS2)
 - First attempt to include small geometry effects
 - Inclusion of the channel-bulk depletion charge results in the familiar $3/2$ power terms
 - Introduced a simple subthreshold model which was not continuous with the strong inversion model.
 - Model became quite complicated and probably is best known as a “developing ground” for better modeling techniques.
- Level 3 (MOS3)
 - Used to overcome the limitations of Level 2. Made use of a semi-empirical approach.
 - Added DIBL and the reduction of mobility by the lateral field.
 - Similar to Level 2 but considerably more efficient.
 - Used binning but was poorly implemented.

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INTRODUCTION TO MODELING

Second Generation Model

- BSIM (Berkeley Short-Channel IGFET Model)
 - Emphasis is on mathematical conditioning for circuit simulation
 - Short channel models are mostly empirical and shifts the modeling to the parameter extraction capability
 - Introduced a more detailed subthreshold current model with good continuity
 - Poor modeling of channel conductance
- HSPICE Level 28
 - Based on BSIM but has been extensively modified.
 - More suitable for analog circuit design
 - Uses model binning
 - Model parameter set is almost entirely empirical
 - User is locked into HSPICE
 - Model is proprietary
- BSIM2
 - Closely based on BSIM
 - Employs several expressions developed from two dimensional analysis
 - Makes extensive modifications to the BSIM model for mobility and the drain current
 - Uses a new subthreshold model
 - Output conductance model makes the model very suitable for analog circuit design

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INTRODUCTION TO MODELING

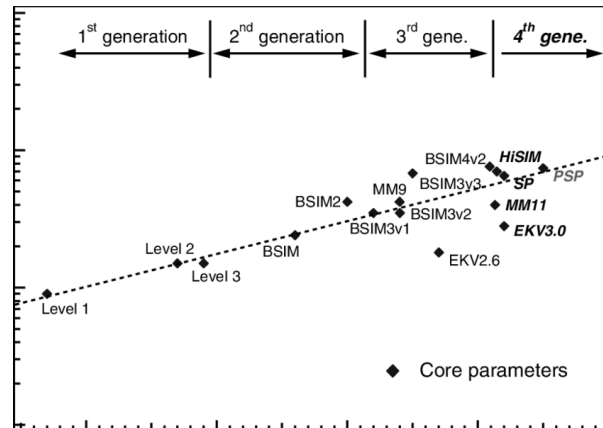
Third Generation Model

- BSIM2 – Continued
 - The drain current model is more accurate and provides better convergence
 - Becomes more complex with a large number of parameters
 - No provisions for variations in the operating temperature
- BSIM3
 - This model has achieved stability and is being widely used in industry for deep submicron technology.
 - Initial focus of simplicity was not realized.
- MOS Model 9
 - Developed at Philips Laboratory
 - Has extensive heritage of industrial use
 - Model equations are clean and simple – should be efficient
- EKV
 - EKV (Enz-Krummenacher-Vittoz) – fresh approach well suited to the needs of analog circuit design

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INTRODUCTION TO MODELING

Summary of Model Generations



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INTRODUCTION TO MODELING

Mô hình thể hệ thứ 1

- Mang nặng tính Vật lý hơn mô phỏng, mô tả đơn giản.
- Thể hệ đầu tiên dựa trên cơ sở vật lý với tất cả các phụ thuộc hình học được gộp vào trong các phương trình mô hình.
- Các mô hình này tập trung vào sự mô tả FET hơn là đặc trưng của các phương trình mô hình trong một bộ mô phỏng mạch.
- Thể hệ của các mô hình này gồm Mức 1, Mức 2 và Mức 3.
- Mức 1: lỗi thời và chỉ hữu hiệu trong giảng dạy
- Mức 2: gộp các hiệu ứng hình học nhỏ, mô tả chi tiết hơn vùng hiểm, điện thế ngưỡng và suy giảm độ di động bởi điện trường ngang. Tuy nhiên kết quả mô phỏng mạch chậm, không hiệu quả
- Mức 3: sử dụng phương pháp bán thực nghiệm, phân đoạn mô hình, hiệu quả hơn Mức 2. Mô hình tương đối đơn giản và cứng cáp. Dùng trong thiết kế mạch chữ số, không phù hợp thiết kế mạch tương tự.

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INTRODUCTION TO MODELING

Mô hình thể hệ thứ 2

- Thể hệ thứ hai thay đổi lớn so với thể hệ đầu tiên. Kết nối giữa vật lý và thực nghiệm, tập trung trọn vẹn vào ứng dụng mô phỏng mạch. Các hàm số đa thức được sử dụng nhiều
- Tham số cấu trúc gồm hai phần: tham số linh kiện cá biệt và tham số hình học (độ dài và độ rộng kênh). Cấu trúc mô hình và các tham số của nó có đặc trưng thực nghiệm đáng kể
- Các mô hình thuộc thể hệ này gồm BSIM, BSIM2 và HSPICE Mức 28 (được dựa trên BSIM).
- BSIM: tập trung vào mô phỏng mạch, cải thiện nhiều cấu trúc toán học của mô hình (như điện thế ngưỡng, độ di động, điện thế bão hòa,...). Đây là mô hình cho thiết kế mạch chữ số được cải thiện và không phù hợp cho thiết kế mạch tương tự. Không chính xác ở độ dài kênh dưới micron
- HSPICE Mức 28: của công ty Meta-Software, dựa trên BSIM nhưng đã được bổ chính cho phù hợp với thiết kế mạch tương tự. Các tham số mô hình hoàn toàn thực nghiệm và dùng phổ biến trong công nghiệp
- BSIM2: sát với BSIM. Nhưng có bổ chính (độ di động, dòng điện máng, độ dẫn ngõ ra,...). Phù hợp với thiết kế mạch tương tự. Tuy nhiên, BSIM2 phức tạp vì chứa nhiều tham số

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INTRODUCTION TO MODELING

Mô hình thể hệ thứ 3

- Các mô hình thuộc thể hệ thứ ba có cấu trúc mô hình đơn giản hơn và số lượng tham số giảm
- Các tham số được dựa trên cơ sở vật lý hơn là thực nghiệm.
- Các mô hình thuộc thể hệ thứ ba sử dụng các hàm làm trơn và chỉ một phương trình được dùng cho tất cả các vùng hoạt động của linh kiện.
- Các mô hình thuộc thể hệ này gồm BSIM3 và MOS Mức 9.
- BSIM3: khắc phục nhược điểm của phiên bản 1 và 2 dựa trên các phương trình thực nghiệm và các tham số mô hình thực nghiệm mới. Tuy nhiên, mô hình và việc trích tham số sẽ phức tạp.
- MOS Mức 9: phát triển bởi Viện Phillips, mô hình không xuất phát từ UCB được sử dụng rộng rãi trong công nghiệp, phương trình mô hình khá gọn và đơn giản nên quá trình mô phỏng mạch và trích tham số sẽ đơn giản. Các hàm hyperbol làm trơn được sử dụng và số lượng tham số khá ít.

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INTRODUCTION TO MODELING

Why modeling?

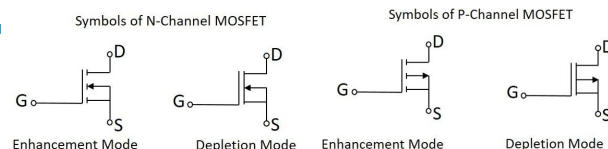
- A model is a kind of theoretical descriptions.
- In an ideal world, we would have a model that accurately describes the behavior of a device under all possible conditions.
- Realistically, we have a model which can predict simulated performance to within a few percent of measured performance.
- There's a gap between theoretical models vs. empirical models. We must improve and modify the theoretical models to describe the behaviors in more details and to be close to the empirical models.
- The model is the closest to becoming a standard for circuit simulators (such as SPICE) on computer.

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MOS TRANSISTOR

Device Structure

- MOSFETs also have three terminals, namely Drain (D), Source (S) and Gate (G) and also one more (optional) terminal called substrate or Body (B). MOSFETs are also available in both types, N-channel (NMOS) and P-channel (PMOS). MOSFETs are basically classified into two forms. They are:
 - Depletion Type
 - Enhancement Type
- **Depletion type:** The depletion type MOSFET transistor is equivalent to a “**Normally Closed**” switch. The depletion type of transistors requires gate – source voltage (V_{GS}) to switch **OFF** the device. The continuous thick line connected between the drain and source terminal represents the depletion type.
- **Enhancement type:** The enhancement type MOSFET transistor is equivalent to a “**Normally Open**” switch and these types of transistors require a gate-source voltage to switch **ON** the device. We can observe that a broken line is connected between the source and drain, which represents the enhancement mode type.
 - In enhancement mode MOSFETs, the conductivity increases by increasing the oxide layer, which adds the carriers to the channel.
 - Generally, this oxide layer is called as ‘Inversion layer’. The channel is formed between the drain and source in the opposite type to the substrate, such as N-channel is made with a P-type substrate and P-channel is made with an N-type substrate. The conductivity of the channel due to electrons or holes depends on N-type or P-type channel respectively.

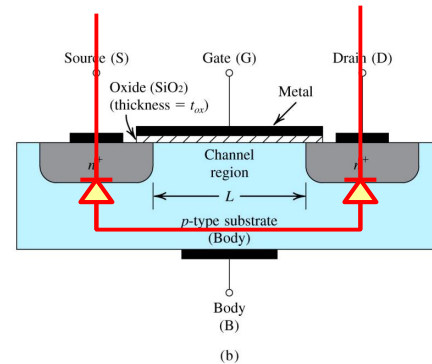


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MOS TRANSISTOR

Device Structure

- With zero voltage applied to gate, **two back-to-back diodes** exist in series between drain and source.
- "They" **prevent current conduction (no conduction)** from drain to source when a voltage v_{DS} is applied.
 - yielding very high resistance (10^{12} **ohms**)

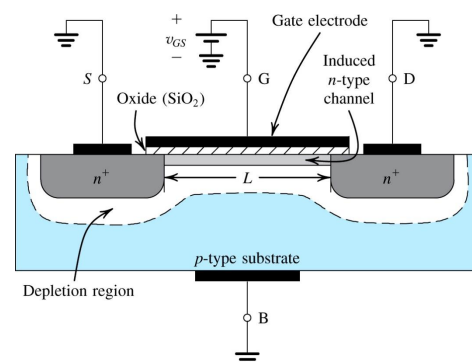


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MOS TRANSISTOR

Device Structure

- What happens if (1) **source and drain are grounded** and (2) **positive voltage is applied to gate**? Refer to figure to right.
- step #1:** v_{GS} is applied to the gate terminal, causing a **positive build up of positive charge** along metal electrode.
- step #2:** This "build up" causes **free holes to be repelled** from region of p -type substrate under gate.



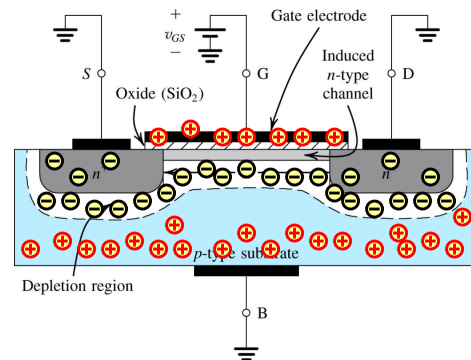
The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate

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MOS TRANSISTOR

Device Structure

- What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.
- step #3:** This "migration" results in the uncovering of negative **bound charges**, originally neutralized by the free holes
- step #4:** The positive gate voltage also **attracts electrons** from the n^+ source and drain regions into the channel.



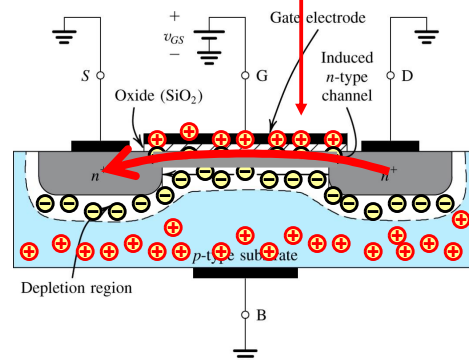
The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate

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MOS TRANSISTOR

Device Structure

- Q:** What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.
- step #5:** Once a sufficient number of "these" electrons accumulate, an n -region is created...
 - ...connecting the source and drain regions
- step #6:** This **provides path for current (conduction)** flow between D and S .
- Inversion layer:** n -channel is formed under the gate oxide. It is inverted from p -type to n -type due to electron accumulation.



The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate

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SIMPLE LARGE SIGNAL MODEL (SAH MODEL)

Large Signal Model Derivation

1. Let the charge per unit area in the channel inversion layer be

$$Q_I(y) = -C_{ox}[v_{GS} - v(y) - V_T] \quad (\text{coul./cm}^2)$$

2. Define sheet conductivity of the inversion layer per square as

$$\sigma_S = \mu_o Q_I(y) \left(\frac{\text{cm}^2}{\text{v} \cdot \text{s}} \right) \left(\frac{\text{coulombs}}{\text{cm}^2} \right) = \frac{\text{amps}}{\text{volt}} = \frac{1}{\Omega/\text{sq.}}$$

3. Ohm's Law for current in a sheet is

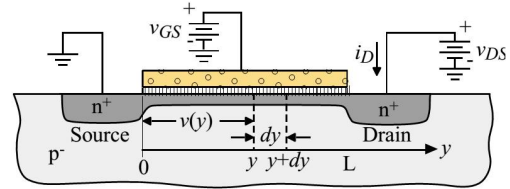
$$J_S = \frac{i_D}{W} = -\sigma_S E_y = -\sigma_S \frac{dv}{dy} \rightarrow dv = \frac{-i_D}{\sigma_S W} dy = \frac{-i_D dy}{\mu_o Q_I(y) W} \rightarrow i_D dy = -W \mu_o Q_I(y) dv$$

4. Integrating along the channel for 0 to L gives

$$\int_0^L i_D dy = - \int_0^{v_{DS}} W \mu_o Q_I(y) dv = \int_0^{v_{DS}} W \mu_o C_{ox} [v_{GS} - v(y) - V_T] dv$$

5. Evaluating the limits gives

$$i_D = \frac{W \mu_o C_{ox}}{L} \left[(v_{GS} - V_T) v(y) - \frac{v^2(y)}{2} \right]_0^{v_{DS}} \rightarrow i_D = \frac{W \mu_o C_{ox}}{L} \left[(v_{GS} - V_T) v_{DS} - \frac{v_{DS}^2}{2} \right]$$



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SIMPLE LARGE SIGNAL MODEL (SAH MODEL)

Simple MOS Large-Signal Model (SPICE MOS Level 1)

- I-V equation:

$$i_D = \frac{\mu_o C_{ox} W}{L} \left[(v_{GS} - v_T) - \frac{v_{DS}}{2} \right] v_{DS}$$

$$i_D = \beta \left[(v_{GS} - v_T) - \frac{v_{DS}}{2} \right] v_{DS} = K' \frac{W}{L} \left[(v_{GS} - v_T) - \frac{v_{DS}}{2} \right] v_{DS}$$

$$\beta = K' \frac{W}{L} \cong \mu_o C_{ox} \frac{W}{L}$$

- μ_o = surface mobility of the channel for the n-channel or p-channel ($\text{cm}^2/\text{V} \cdot \text{s}$)
- W = effective channel width
- L = effective channel length
- Gate oxide capacitance C_{ox} : capacitance per unit area of the gate oxide (F/cm^2)

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \epsilon_0}{t_{ox}}$$

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SIMPLE LARGE SIGNAL MODEL (SAH MODEL)

Simple MOS Large-Signal Model (SPICE MOS Level 1)

- Threshold voltage V_T :

$$V_T = V_{T0} + \gamma \left(\sqrt{2|\phi_F| + v_{SB}} - \sqrt{2|\phi_F|} \right)$$

$$V_{T0} = V_T(v_{SB} = 0) = V_{FB} + 2|\phi_F| + \frac{\sqrt{2q\epsilon_{Si}N_{SUB}2|\phi_F|}}{C_{ox}}$$

- Bulk threshold parameter ($V^{1/2}$): $\gamma = \frac{\sqrt{2q\epsilon_{Si}N_{SUB}}}{C_{ox}}$

- Strong inversion surface potential (V): $\phi_F = \frac{kT}{q} \ln\left(\frac{N_{SUB}}{n_i}\right)$ T = Temperature (K)

$$\phi_F(\text{substrate}) = \frac{kT}{q} \ln\left(\frac{N_{SUB}}{n_i}\right) \quad [\text{n-channel with p-substrate}] \quad \phi_F(\text{gate}) = \frac{kT}{q} \ln\left(\frac{N_{GATE}}{n_i}\right) \quad [\text{n-channel with n+ polysilicon gate}]$$

- Flatband voltage (V): $v_{FB} = \phi_{MS} - \frac{Q_{ss}}{C_{ox}}$ (Q_{ss} = oxide-charge = qN_{ss})

- $\phi_{MS} = \phi_F(\text{substrate}) - \phi_F(\text{gate})$

SPICE variable	Equation
TOX	$TOX = t_{ox}$
KP	$KP = \mu' C_{ox}$
VTO	$VTO = V_{FB} + 2\phi_F + \frac{\sqrt{2\epsilon_{Si}qN_d(2\phi_F)}}{C_{ox}}$
GAMMA	$GAMMA = \gamma = \frac{\sqrt{2\epsilon_{Si}qN_d}}{C_{ox}}$
NSUB	$NSUB = N_d \text{ or } N_a$
U0	$U0 = \mu'$
LAMBDA	$LAMBDA = \lambda$
VMAX	$VMAX = v_{sat}$

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SIMPLE LARGE SIGNAL MODEL (SAH MODEL)

The Simple Large Signal MOSFET Model

- Cutoff Region:

$$v_{GS} - V_T < 0$$

$$i_D = 0$$

(Ignores subthreshold currents)

- Triode or Active Region

$$0 < v_{DS} < v_{GS} - V_T$$

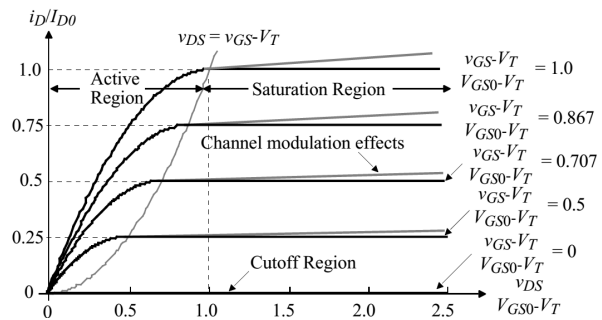
$$i_D = \frac{\mu_0 C_{ox} W}{L} [2(v_{GS} - V_T) - v_{DS}] v_{DS}$$

- Saturation Region

$$0 < v_{GS} - V_T < v_{DS}$$

$$i_D = \frac{\mu_0 C_{ox} W}{L} (v_{GS} - V_T)^2 \quad \therefore i_D = \frac{\mu_0 C_{ox} W}{L} (v_{GS} - V_T)^2 (1 + \lambda v_{DS})$$

Output Characteristics of the MOSFET

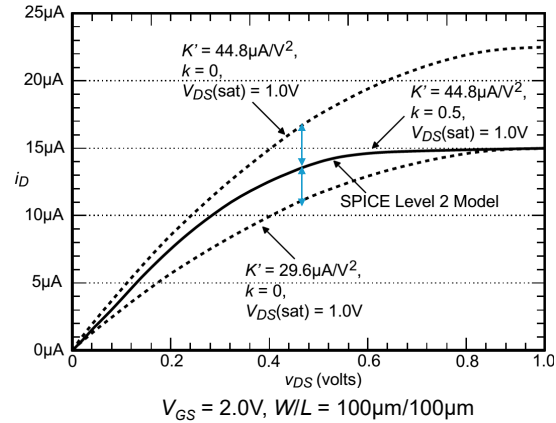


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SIMPLE LARGE SIGNAL MODEL (SAH MODEL)

Performance Limitations of the Simple Sah

- If we compare the Sah model to a more precise model (SPICE level 2) that the Sah model has issues with the “knee” area as shown.



SAH Model Modification

- This difference is due to the fact that we assumed that the threshold, V_T , was constant over the channel.
- If we let $V_T(y) = V_T + kv(y)$ then the Sah model is exactly the same as the SPICE model.

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SIMPLE LARGE SIGNAL MODEL (SAH MODEL)

Modification of the Previous Model to Include the Effects of v_{DS} on V_T

- From the previous derivation:

$$\int_0^L i_D dy = - \int_0^{v_{DS}} W \mu_o Q(y) dv = \int_0^{v_{DS}} W \mu_o C_{ox} [v_{GS} - v(y) - V_T] dv$$
- Assume that the threshold voltage varies across the channel in the following way:

$$V_T(y) = V_T + kv(y)$$

where V_T is the value of V_T at the source end of the channel and k is a constant.

- Integrating the above gives,

$$i_D = \frac{W \mu_o C_{ox}}{L} \left[(v_{GS} - V_T) v(y) - (1+k) \frac{v(y)^2}{2} \right]_0^{v_{DS}} \rightarrow i_D = \frac{W \mu_o C_{ox}}{L} \left[(v_{GS} - V_T) v_{DS} - (1+k) \frac{v_{DS}^2}{2} \right]$$

- To find $v_{DS}(\text{sat})$, set the di_D/dv_{DS} equal to zero and solve for $v_{DS} = v_{DS}(\text{sat})$,

$$v_{DS}(\text{sat}) = \frac{v_{GS} - V_T}{1+k}$$

- Therefore, in the saturation region, the drain current is

$$i_D = \frac{W \mu_o C_{ox}}{2(1+k)L} (v_{GS} - V_T)^2$$

- For $k = 0.5$ and $K' = 44.8 \mu\text{A}/\text{V}^2$, excellent correlation is achieved with SPICE 2 as seen on the previous slide.

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SIMPLE LARGE SIGNAL MODEL (SAH MODEL)

Simple MOS Large-Signal Model (SPICE MOS Level 1)

- Constants for Silicon

Constant Symbol	Constant Description	Value	Units
V_{GO}	Silicon bandgap (27°C)	1.205	V
k	Boltzmann's constant	1.381×10^{-23}	J/K
n_i	Intrinsic carrier concentration (27°C)	1.45×10^{10}	cm^{-3}
ϵ_0	Permittivity of free space	8.854×10^{-14}	F/cm
ϵ_{Si}	Permittivity of Silicon	$11.7 \epsilon_0$	F/cm
ϵ_{ox}	Permittivity of SiO_2	$3.9 \epsilon_0$	F/cm

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SIMPLE LARGE SIGNAL MODEL (SAH MODEL)

MOSFET Parameters

Model Parameters for a Typical CMOS Bulk Process (0.25 μm CMOS n-well):

Parameter Symbol	Parameter Description	Typical Parameter Value		Units
		N-Channel	P-Channel	
V_{T0}	Threshold Voltage ($V_{\text{BS}} = 0$)	0.5 ± 0.15	-0.5 ± 0.15	V
K'	Transconductance Parameter (in saturation)	$120.0 \pm 10\%$	$25.0 \pm 10\%$	$\mu\text{A}/\text{V}^2$
γ	Bulk threshold parameter	0.4	0.6	$(\text{V})^{1/2}$
λ	Channel length modulation parameter	$0.32 (L=L_{\text{min}})$ $0.06 (L \geq 2L_{\text{min}})$	$0.56 (L=L_{\text{min}})$ $0.08 (L \geq 2L_{\text{min}})$	$(\text{V})^{-1}$
$2 \phi_F $	Surface potential at strong inversion	0.7	0.8	V

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SIMPLE LARGE SIGNAL MODEL (SAH MODEL)

Summary of the Simple Large Signal MOSFET Model

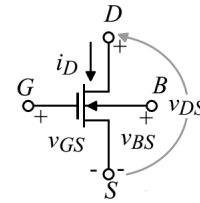
- N-channel reference convention (Terms in **red** are model parameters):

- Non-saturation:
$$i_D = \frac{W\mu_0 C_{ox}}{L} \left[(v_{GS} - V_T)v_{DS} - \frac{v_{DS}^2}{2} \right] (1 + \lambda v_{DS})$$

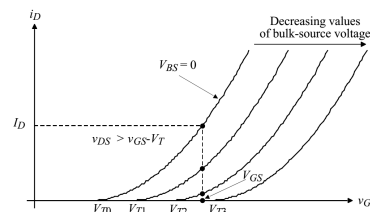
- Saturation:
$$i_D = \frac{W\mu_0 C_{ox}}{L} \left[(v_{GS} - V_T)v_{DS(sat)} - \frac{v_{DS(sat)}^2}{2} \right] (1 + \lambda v_{DS}) = \frac{W\mu_0 C_{ox}}{2L} (v_{GS} - V_T)^2 (1 + \lambda v_{DS})$$

where

- μ_0 = zero field mobility (cm²/volt-sec)
- C_{ox} = gate oxide capacitance per unit area (F/cm²)
- λ = channel-length modulation parameter (volts⁻¹)
- $V_T = V_{T0} + \gamma \left(\sqrt{2|\phi_f| - v_{BS}} - \sqrt{2|\phi_f|} \right)$ = threshold voltage
- V_{T0} = zero bias threshold voltage
- γ = bulk threshold parameter (volts^{0.5})
- $2|\phi_f|$ = strong inversion surface potential (volts)
- For p-channel MOSFETs, use n-channel equations with p-channel parameters and invert the current.



Bulk-Source (v_{BS}) influence on the transconductance characteristics and threshold voltage V_T , v_{BS} increases $\rightarrow V_T$ increases



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INTRODUCTION TO MODELING

MOSFET Models in SPICE Simulators

MOS LEVEL 1 NMOS Parameters

```
.MODEL L1 NMOS LEVEL=1 RSH=0 TOX=300E-10 LD=0.21E-6 XJ=0.3E-6 VMAX=15E4
ETA=0.18 GAMMA=0.4 KAPPA=0.5+ NSUB=35E14 UO=700 THETA=0.095 VTO=0.781
CGSO=2.8E-10+ CGDO=2.8E-10 CJ=5.75E-5 CJSW=2.48E-10 PB=0.7 MJ=0.5+ MJSW=0.3
NFS=1E10
```

MOS LEVEL 2 NMOS Parameters

```
.MODEL L2NMOS LEVEL=2 PHI=0.7 TOX=4.08E-8 XJ=0.2 TPG=1 +VTO=0.8096 DELTA=4.586
LD=2.972E-7 KP=5.3532E-5 +UO=632.5 UEXP=0.1328 UCRIT=3.897E4 RSH=6.202
+GAMMA=0.5263 NSUB=5.977E15 NFS=5.925E11 VMAX=5.83E4+LAMBDA=3.903E-2
CGDO=3.7731E-10 CGSO=3.7731E-10 +CGB0=3.4581E-10 CJ=1.3679E-4 MJ=0.63238
CJSW=5.1553E-10 +MJSW=0.26805 PB=0.4
```

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INTRODUCTION TO MODELING

MOSIS Parametric Test Results

<http://www.mosis.org/>

RUN: T02D VENDOR: TSMC
TECHNOLOGY: SCN025 FEATURE SIZE: 0.25 microns

INTRODUCTION: This report contains the lot average results obtained by MOSIS from measurements of MOSIS test structures on each wafer of this fabrication lot. SPICE parameters obtained from similar measurements on a selected wafer are also attached.

COMMENTS: TSMC 0251P5M.

TRANSISTOR PARAMETERS	W/L	N-CHANNEL	P-CHANNEL	UNITS
MINIMUM	0.36/0.24			
Vth		0.54	-0.50	volts
SHORT	20.0/0.24			
Idss		557	-256	uA/um
Vth		0.56	-0.56	volts
Vpt		7.6	-7.2	volts
WIDE	20.0/0.24			
Ids0		6.6	-1.5	pA/um
LARGE	50.0/50.0			
Vth		0.47	-0.60	volts
Vjbkd		5.8	-7.0	volts
Ijlk		-25.0	-1.1	pA
Gamma		0.44	0.61	V ^{0.5}
K' (Uo*Cox/2)		112.0	-23.0	uA/V ²

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INTRODUCTION TO MODELING

0.25µm BSIM3v3.1 NMOS Parameters

.MODEL CMOSN NMOS (LEVEL = 49

+VERSION = 3.1 TNOM = 27 TOX = 5.7E-9 XJ = 1E-7 NCH = 2.3549E17 VTH0 = 0.4273342
+K1 = 0.3922983 K2 = 0.0185825 K3 = 1E-3 K3B = 2.0947677 W0 = 2.171779E-7 NLX = 1.919758E-7
+DVT0W = 0 DVT1W = 0 DVT2W = 0 DVT0 = 7.137212E-3 DVT1 = 6.066487E-3 DVT2 = -0.3025397
+U0 = 403.1776038 UA = -3.60743E-12 UB = 1.323051E-18 UC = 2.575123E-11 VSAT = 1.616298E5 A0 = 1.4626549
+AGS = 0.3136349 B0 = 3.080869E-8 B1 = -1E-7 KETA = 5.462411E-3 A1 = 4.653219E-4 A2 = 0.6191129
+RDSW = 345.624986 PRWG = 0.3183394 PRWB = -0.1441065 WR = 1 WINT = 8.107812E-9 LINT = 3.375523E-9
+XL = 3E-8 XW = 0 DWG = 6.420502E-10 DWB = 1.042094E-8 VOFF = -0.1083577 NFACTOR = 1.1884386
+CIT = 0 CDSC = 2.4E-4 CDSCD = 0 CDSCB = 0 ETA0 = 4.914545E-3 ETAB = 4.215338E-4
+DSUB = 0.0313287 PCLM = 1.2088426 PDIBLC1 = 0.7240447 PDIBLC2 = 5.120303E-3 PDIBLCB = -0.0443076 DROUT = 0.7752992
+PSCBE1 = 4.451333E8 PSCBE2 = 5E-10 PVAG = 0.2068286 DELTA = 0.01 MOBMOD = 1 PRT = 0
+UTE = -1.5 KT1 = -0.11 KT1L = 0 KT2 = 0.022 UA1 = 4.31E-9 UB1 = -7.61E-18
+UC1 = -5.6E-11 AT = 3.3E4 WL = 0 WLN = 1 WW = -1.22182E-16 WWN = 1.2127
+WWL = 0 LL = 0 LLN = 1 LW = 0 LWN = 1 LWL = 0
+CAPMOD = 2 XPART = 0.4 CGDO = 6.33E-10 CGSO = 6.33E-10 CGBO = 1E-11 CJ = 1.766171E-3
+PB = 0.9577677 MJ = 0.4579102 CJSW = 3.931544E-10 PBSW = 0.99 MJSW = 0.2722644 CF = 0
+PVTH0 = -2.126483E-3 PRDSW = -24.2435379 PK2 = -4.788094E-4 WKETA = 1.430792E-3 LKETA = -6.548592E-3)

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INTRODUCTION TO MODELING

0.25 μ m BSIM3v3.1 PMOS Parameters

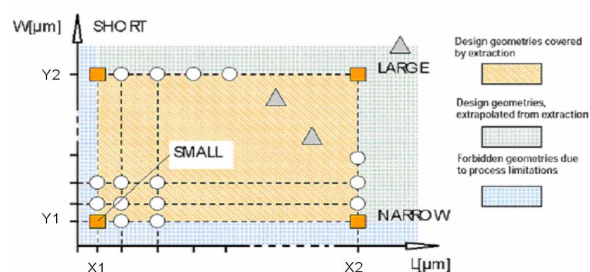
```
.MODEL CMOSP PMOS ( LEVEL = 49
+VERSION = 3.1 TNOM = 27 TOX = 5.7E-9 XJ = 1E-7 NCH = 4.1589E17 VTH0 = -0.6193382
+K1 = 0.5275326 K2 = 0.0281819 K3 = 0 K3B = 11.249555 W0 = 1E-6 NLX = 1E-9
+DVT0W = 0 DVT1W = 0 DVT2W = 0 DVT0 = 3.1920483 DVT1 = 0.4901788 DVT2 = -0.0295257
+U0 = 185.1288894 UA = 3.40616E-9 UB = 3.640498E-20 UC = -6.35238E-11 VSAT = 1.975064E5 A0 = 0.4156696
+AGS = 0.0702036 B0 = 3.111154E-6 B1 = 5E-6 KETA = 0.0253118 A1 = 2.421043E-4 A2 = 0.6754231
+RDSW = 866.896668 PRWG = 0.0362726 PRWB = -0.293946 WR = 1 WINT = 6.519911E-9 LINT = 2.210804E-8
+XL = 3E-8 XW = 0 DWG = -2.423118E-8 DWB = 3.052612E-8 VOFF = -0.1161062 NFACTOR = 1.2546896
+CIT = 0 CDSC = 2.4E-4 CDSCD = 0 CDSCB = 0 ETA0 = 0.7241245 ETAB = -0.3675267
+DSUB = 1.1734643 PCLM = 1.0837457 PDIBLC1 = 9.608442E-4 PDIBLC2 = 0.0176785 PDIBLCB = -9.605935E-4 DROUT = 0.0735541
+PSCBE1 = 1.579442E10 PSCBE2 = 6.707105E-9 PVAG = 0.0409261 DELTA = 0.01 MOBMOD = 1 PRT = 0
+UTE = -1.5 KT1 = -0.11 KT1L = 0 KT2 = 0.022 UA1 = 4.31E-9 UB1 = -7.61E-18
+UC1 = -5.6E-11 AT = 3.3E4 WL = 0 WLN = 1 WW = 0 WWN = 1
+WWL = 0 LL = 0 LLN = 1 LW = 0 LWN = 1 LWL = 0 CAPMOD = 2 XPART = 0.4 CGDO = 5.11E-10
+CGSO = 5.11E-10 CGBO = 1E-11 CJ = 1.882953E-3 PB = 0.99 MJ = 0.4690946 CJSW = 3.018356E-10
+PBSW = 0.8137064 MJSW = 0.3299497 CF = 0 PVTH0 = 5.268963E-3 PRDSW = -2.2622317 PK2 = 3.952008E-3
+WKETA = -7.69819E-3 LKETA = -0.0119828 )
```

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INTRODUCTION TO MODELING

MOSFET Bining

- Large transistors (W large, L large)
- Short transistors (W large, L small)
- Narrow Transistors (W small, L large)
- Small Transistors (W small, L small)

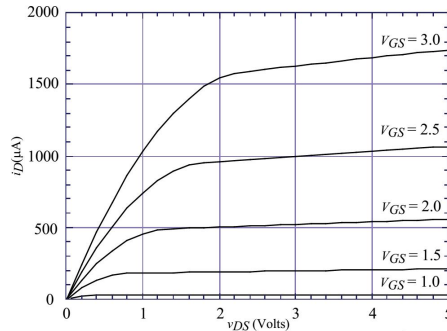


- Narrow and short transistors have Narrow channel effect and Short channel effect, respectively due to transistor's physical sizes.

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OPERATION OF THE MOS TRANSISTOR

Output Characteristics (I_D - V_{DS}) of an Enhancement NMOS Transistor



SPICE Input File:

Output Characteristics for NMOS

M1 6 1 0 0 MOS1 w=5u l=1.0u

VGS1 1 0 1.0

M2 6 2 0 0 MOS1 w=5u l=1.0u

VGS2 2 0 1.5

M3 6 3 0 0 MOS1 w=5u l=1.0u

VGS3 3 0 2.0

M4 6 4 0 0 MOS1 w=5u l=1.0u

VGS4 4 0 2.5

M5 6 5 0 0 MOS1 w=5u l=1.0u

VGS5 5 0 3.0

VDS 6 0 5

.model mos1 nmos (vto=0.7 kp=110u

+gamma=0.4 lambda=.04 phi=.7)

.dc vds 0 5 .2

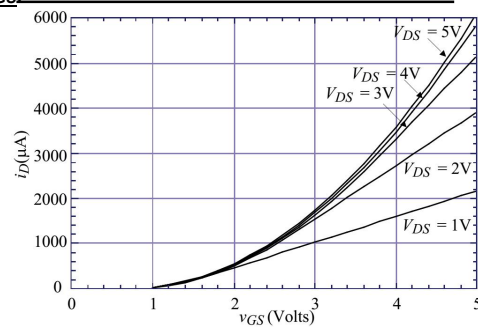
.print dc ID(M1), ID(M2), ID(M3), ID(M4), ID(M5)

.end

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OPERATION OF THE MOS TRANSISTOR

Input Characteristics (I_D - V_{GS}) of an Enhancement NMOS Transistor



SPICE Input File:

Transconductance Characteristics for NMOS

M1 1 6 0 0 MOS1 w=5u l=1.0u

VDS1 1 0 1.0

M2 2 6 0 0 MOS1 w=5u l=1.0u

VDS2 2 0 2.0

M3 3 6 0 0 MOS1 w=5u l=1.0u

VDS3 3 0 3.0

M4 4 6 0 0 MOS1 w=5u l=1.0u

VDS4 4 0 4.0

M5 5 6 0 0 MOS1 w=5u l=1.0u

VDS5 5 0 5.0

VGS 6 0 5

.model mos1 nmos (vto=0.7 kp=110u

+gamma=0.4 lambda=.04 phi=.7)

.dc vgs 0 5 .2

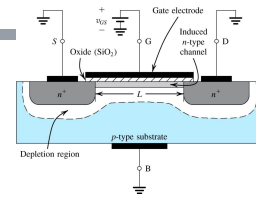
.print dc ID(M1), ID(M2), ID(M3), ID(M4), ID(M5)

.probe

.end

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MOS TRANSISTOR



Channel for Current Flow

- **threshold voltage** (V_t) – is the **minimum value of v_{GS}** required to form a conducting channel between drain and source (**V_{tn} is used for n -type MOSFET, V_{tp} is used for p -channel**)
 - typically between 0.3 and $0.7V_{dc}$
- **field-effect** – when positive v_{GS} is applied, an **electric field** develops between the gate electrode and induced n -channel – the conductivity of this channel is **affected** by the strength of field
 - SiO_2 layer acts as dielectric
- **effective / overdrive voltage** – is the **difference between v_{GS} applied and V_t** .

$$v_{ov} = v_{GS} - V_t$$

- **oxide capacitance** (C_{ox}) – is the **capacitance of the parallel plate capacitor** per unit gate area (F/m^2)

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

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MOS TRANSISTOR

Channel for Current Flow

- Current is defined in terms of **charge per unit length of n -channel** as well as **electron drift velocity**.

μ_n presents mobility of electrons at surface of the n -channel in m^2/Vs

$$i_D = (C_{ox} W v_{ov}) \left(\frac{\mu_n v_{DS}}{L} \right)$$

Charge per unit length of n -channel in C/m Electron drift velocity m^2/Vs

- **For small values of v_{DS}** , the n -channel acts like a **variable resistance** whose value is controlled by v_{ov} .

$$i_D = \left[\mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov} \right] v_{DS}$$

$$r_{DS} = \frac{v_{DS}}{i_D} = \frac{1}{\mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}}$$

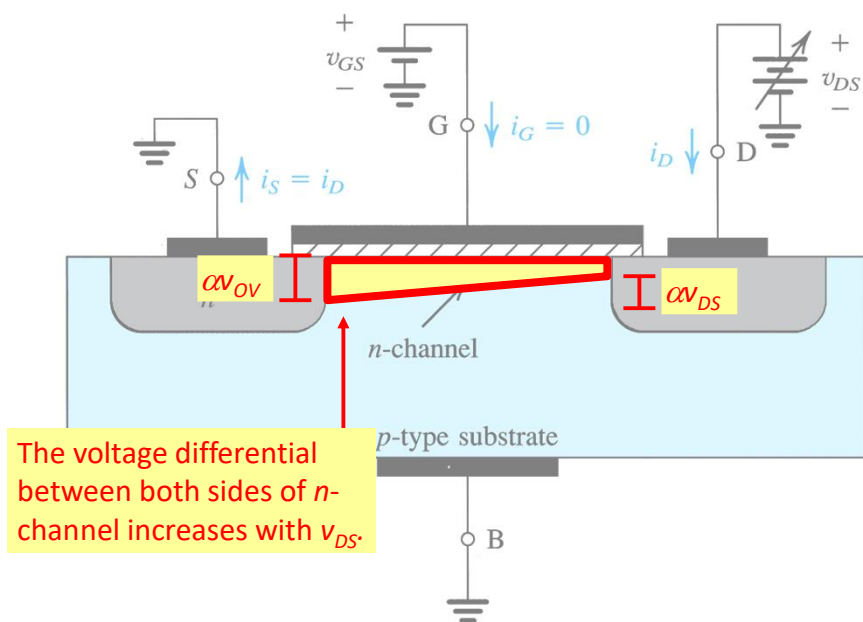
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MOS TRANSISTOR

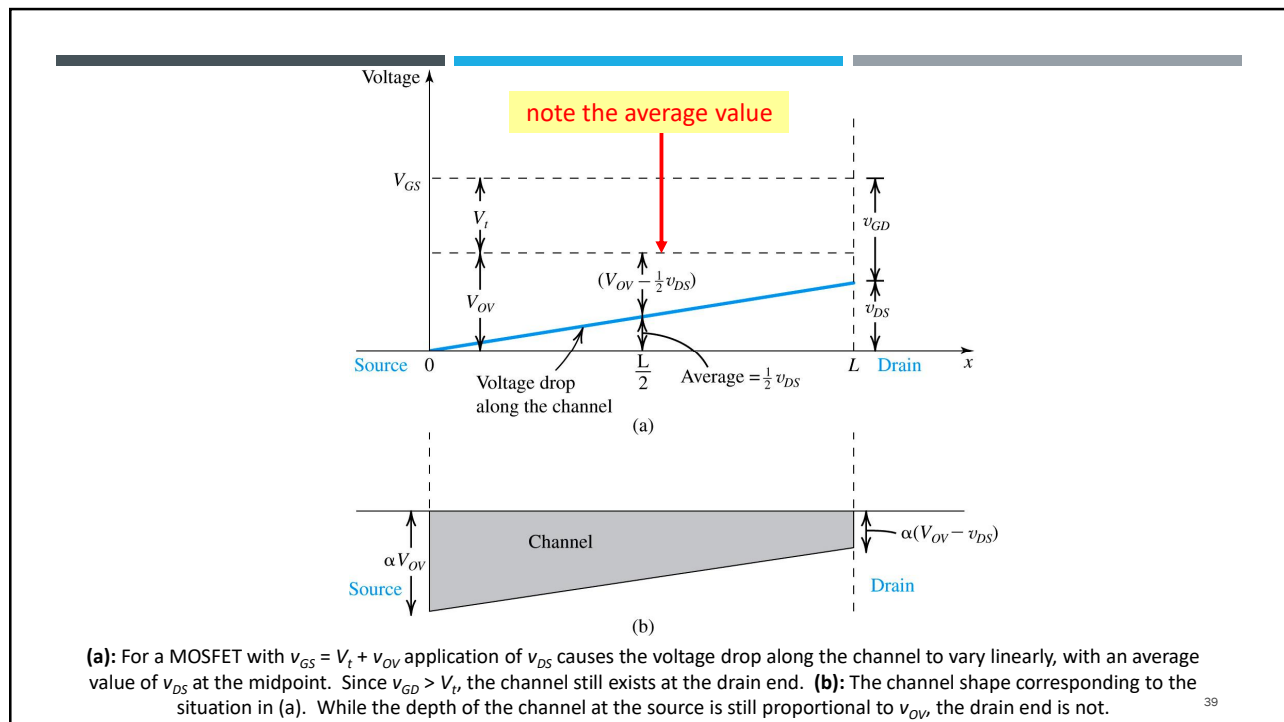
When v_{DS} is increased

- **Step #1:** Assume that v_{GS} is held constant at value greater than V_t .
- **Step #2:** Also assume that v_{DS} is applied and appears as voltage drop across n -channel.
- **Step #3:** Note that voltage decreases from v_{GS} at the source end of channel to v_{GD} at drain end, where...
 - $V_{GD} = V_{GS} - V_{DS}$
 - $V_{GD} = V_t + V_{OV} - V_{DS}$
- Operation of the NMOS transistor as v_{DS} is increased (shown in the next slide)

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MOS TRANSISTOR

When v_{DS} is increased

- Step #4: Define i_{DS} in terms of v_{DS} and v_{OV} .**

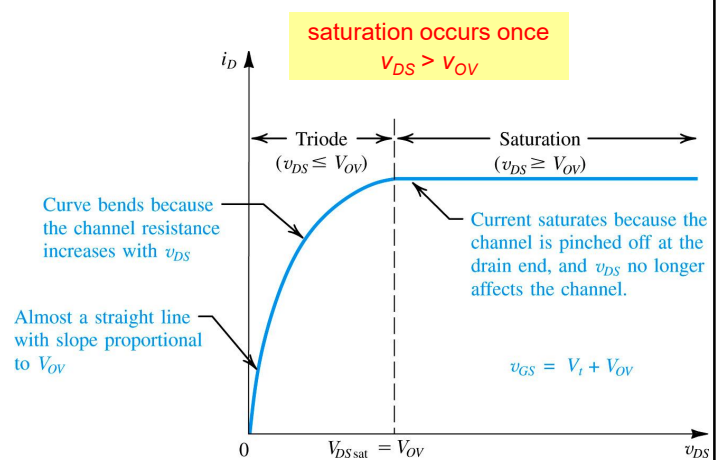
Replace v_{OV} with $v_{OV} - \frac{1}{2}v_{DS}$

- If $v_{DS} < v_{OV}$: (Triode region)

$$i_D = \left[\mu_n C_{ox} \left(\frac{W}{L} \right) \left(v_{OV} - \frac{1}{2}v_{DS} \right) \right] v_{DS}$$

- If $v_{DS} > v_{OV}$: (Saturation region)

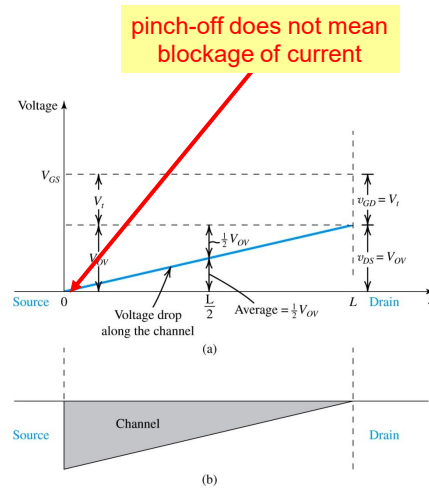
$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{OV}^2$$



MOS TRANSISTOR

When $v_{DS} \gg v_{OV}$

- Operation of MOSFET with $v_{GS} = V_t + v_{OV}$ as v_{DS} is increased to v_{OV} .
- At the drain end, v_{GD} decreases to V_t and the channel depth at the drain-end reduces to zero (pinch-off). At this point, the MOSFET enters saturation mode of operation.
- Further increasing v_{DS} (beyond v_{OV}) has no effect on the channel shape and i_D remains constant.



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MOS TRANSISTOR

Exercise 1:

Consider an NMOS process technology for which $L_{min} = 0.4\mu m$, $t_{ox} = 8nm$, $m_n = 450cm^2/Vs$, $V_t = 0.7V$.

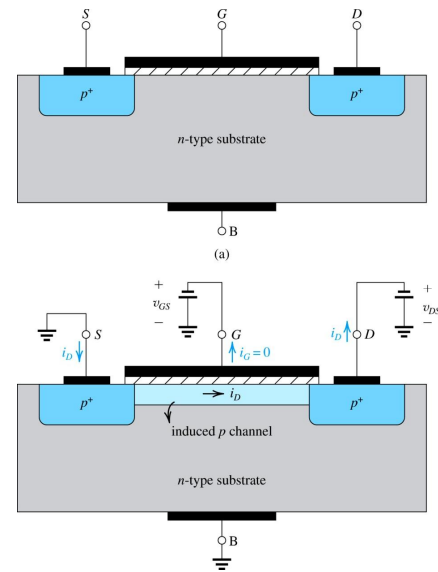
- Find C_{ox} and k'_n .
- For a MOSFET with $W/L = 8\mu m/0.8\mu m$, calculate the values of v_{OV} , v_{GS} , and v_{DSmin} needed to operate the transistor in the saturation region with dc current $I_D = 100\mu A$.
- For the device in (b), find the values of v_{OV} and v_{GS} required to cause the device to operate as a 1000Ω resistor for very small v_{DS} .

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MOS TRANSISTOR

The p-channel MOSFET

- In Figure
- (a): Physical structure of the PMOS transistor. Note that it is similar to the NMOS transistor, except that all semiconductor regions are reversed in polarity.
- (b): A negative voltage v_{GS} of magnitude greater than $|V_{tp}|$ induces a p-channel, and a negative v_{DS} causes a current i_D to flow from source to drain.
- Process transconductance parameters are defined differently
 - $k'_{n/p} = \mu_{n/p} C_{ox}$
 - $k_{n/p} = \mu_{n/p} C_{ox} (W/L)$

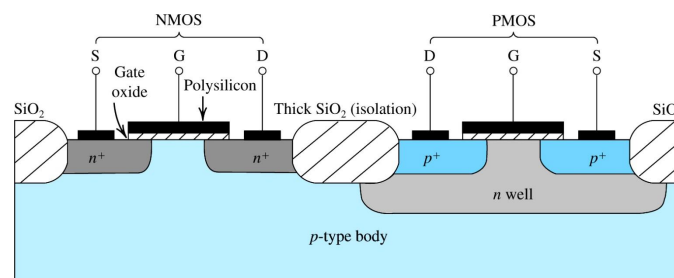


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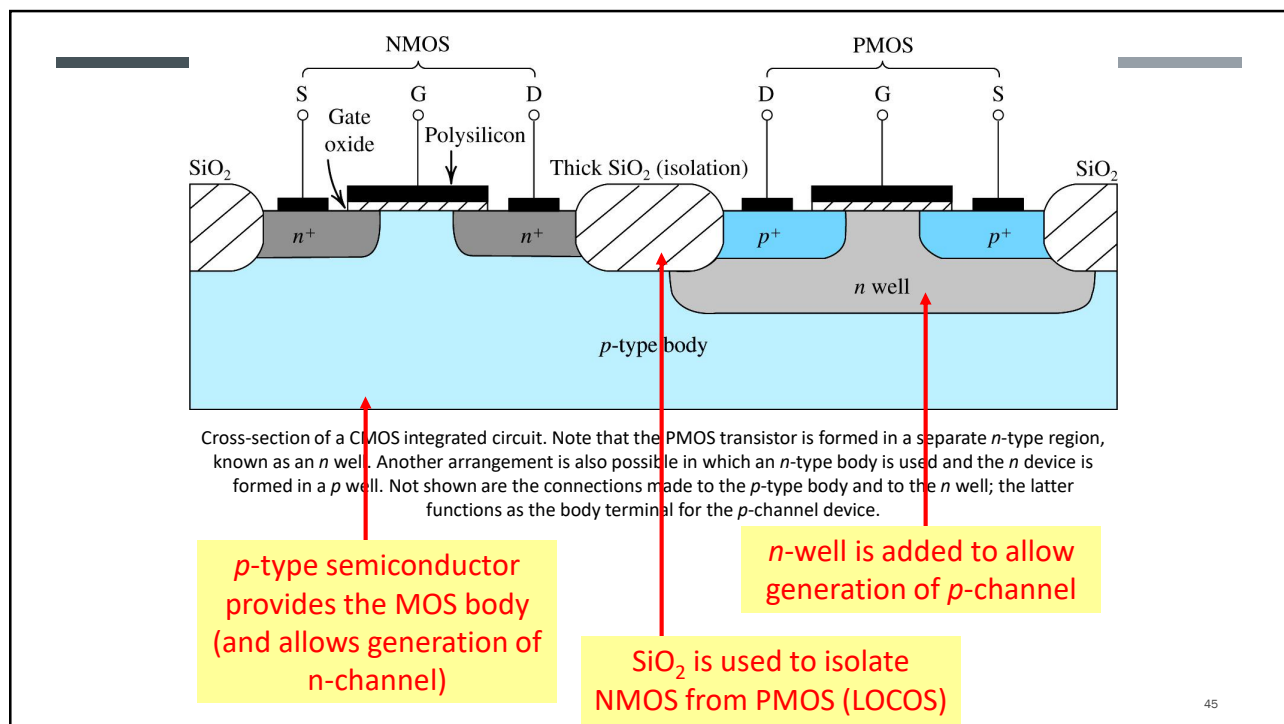
MOS TRANSISTOR

Complementary MOS devices

- complementary devices (CMOS)** – two devices such as the p-channel and n-channel MOSFET's.
 - electron mobility μ_n is 2 – 4 times greater than hole mobility μ_p .
- complementary MOS (CMOS) technology** – is technology which allows fabrication of both NMOS and PMOS transistors on a single chip.



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MOS TRANSISTOR

Recap of i_D

- The equation used to define i_D depends on relationship between v_{DS} and v_{OV} .

- $v_{DS} \ll v_{OV}$

$$i_D = (C_{ox} W v_{ov}) \left(\frac{\mu_n v_{DS}}{L} \right)$$

- $v_{DS} < v_{OV}$

$$i_D = \left[\mu_n C_{ox} \left(\frac{W}{L} \right) \left(v_{ov} - \frac{1}{2} v_{DS} \right) \right] v_{DS}$$

- $v_{DS} \geq v_{OV}$

$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}^2$$

- $v_{DS} \gg v_{OV}$

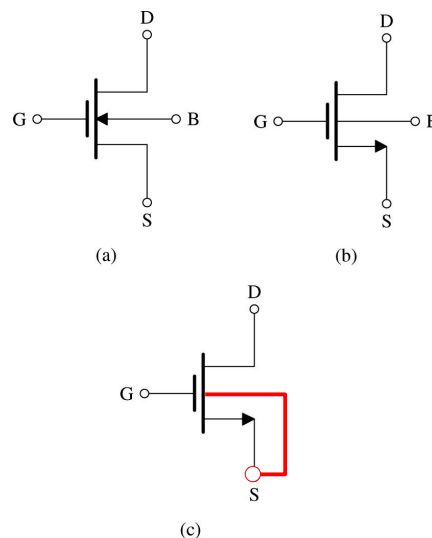
$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}^2 (1 + \lambda v_{DS})$$

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MOS TRANSISTOR

Current-Voltage Characteristics

- The figure shows an n -channel enhancement MOSFET.
- There are **four terminals**:
 - drain (D), gate (G), body (B), and source (S).
- Although, it is **assumed that body and source are connected**.
- (a): Circuit symbol for the n -channel enhancement-type MOSFET.
- (b): Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., n channel).
- (c): Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.



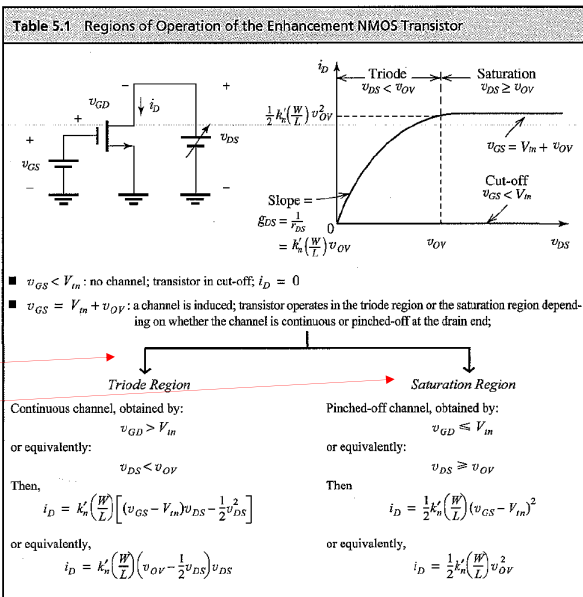
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MOS TRANSISTOR

The i_D - v_{DS} NMOS Characteristics

- The operation of NMOS transistor consists of in three regions.

- Cutoff**
- Triode (or Active)**
- Saturation**



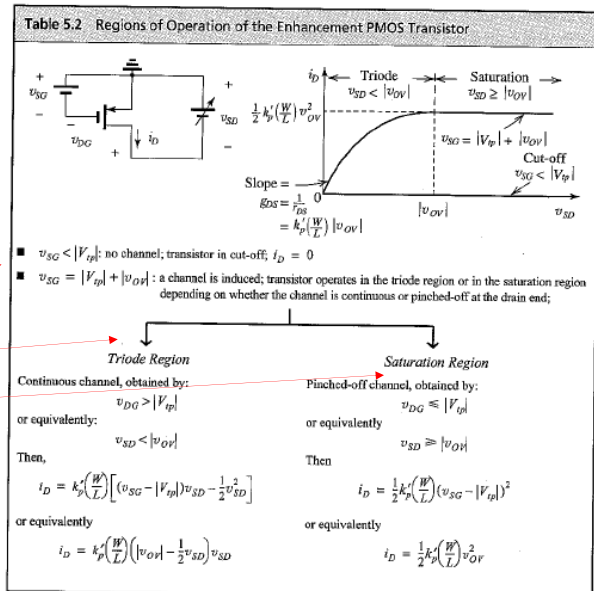
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MOS TRANSISTOR

The i_D - v_{DS} PMOS Characteristics

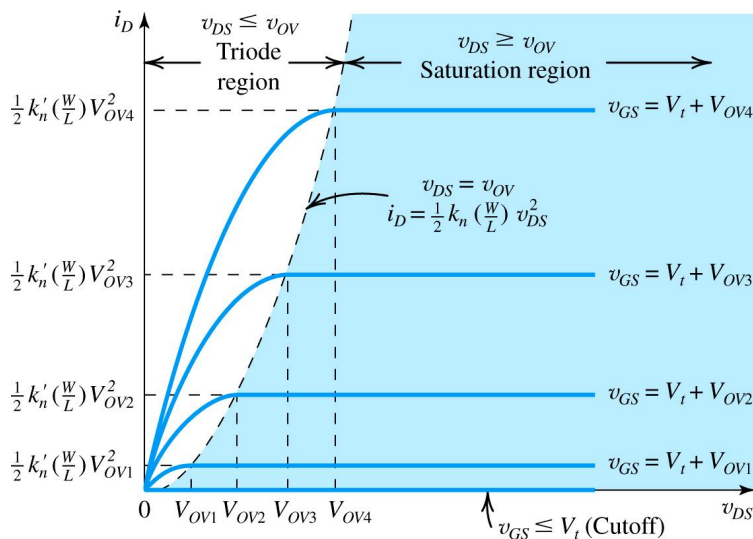
- Characteristics of the PMOS are similar to the NMOS

- Cutoff
- Triode (or Active)
- Saturation



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as v_{GS} increases,
so current i_D
increases and
beginning of the
saturation region



The i_D - v_{DS} characteristics for an enhancement-type NMOS transistor

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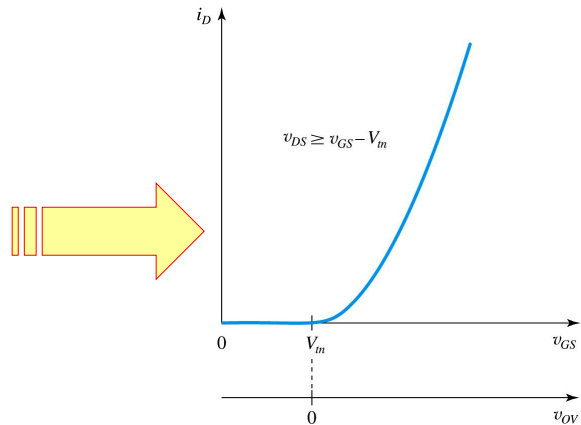
MOS TRANSISTOR

The i_D - v_{GS} Characteristics

- The i_D - v_{GS} characteristic of an NMOS transistor operating in the saturation region. The i_D - v_{OV} characteristic can be obtained by simply re-labeling the horizontal axis, that is, shifting the origin to the point $v_{GS} = V_{tn}$.

$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}^2 = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (v_{GS} - V_{tn})^2$$

- When MOSFET's are employed to design an amplifier, they are operated in saturation region.
- In saturation, the drain current (i_D) is...
 - dependent on v_{GS}
 - independent of v_{DS}
- In effect, it becomes a **voltage-controlled current source (VCCS)**. This is key for amplification.

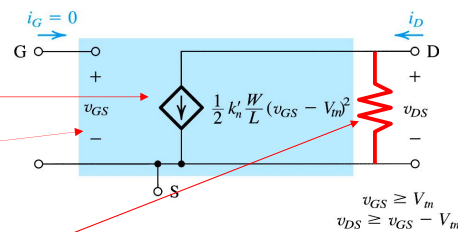


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MOS TRANSISTOR

The i_D - v_{GS} Characteristics

- This circuit is known as the **large-signal equivalent circuit**.
- Current source is **ideal**.
- Infinite output resistance** represents independent, in saturation, of i_D from v_{DS} .



note that, in this circuit, i_D is completely independent of v_{DS} (because no shunt resistor exists)

Large-signal equivalent-circuit model of an n -channel MOSFET operating in the saturation

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MOS TRANSISTOR

Exercise 2:

- Consider an NMOS transistor fabricated in an 0.18- μm process with $L = 0.18\mu\text{m}$ and $W = 2\mu\text{m}$. The process technology is specified to have $C_{ox} = 8.6\text{fF}/\mu\text{m}^2$, $m_n = 450\text{cm}^2/\text{Vs}$, and $V_{tn} = 0.5\text{V}$.
- a) Find V_{GS} and V_{DS} that result in the MOSFET operating at the edge of saturation with $I_D = 100\mu\text{A}$.
- b) If V_{GS} is kept constant, find V_{DS} that results in $I_D = 50\mu\text{A}$.
- c) To investigate the use of the MOSFET as a linear amplifier, let it be operating in saturation with $V_{DS} = 0.3\text{V}$. Find the change in i_D resulting from v_{GS} changing from 0.7V by $+0.01\text{V}$ and -0.01V .

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MOS TRANSISTOR

Finite Output Resistance in Saturation

- In previous section, we assume (in saturation) i_D is independent of v_{DS} .
- Therefore, a change Δv_{DS} causes no change in i_D .
 - This implies that the incremental resistance R_s is infinite.
 - It is based on the idealization that, once the n -channel is pinched off, changes in v_{DS} will have no effect on i_D .
 - The problem is that, in practice, this is not completely true.

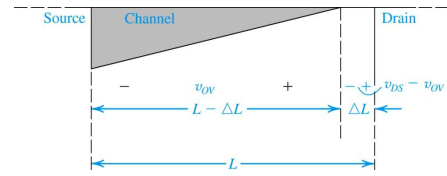
54

MOS TRANSISTOR

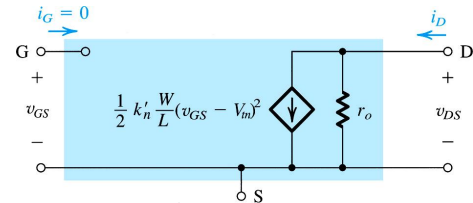
Finite Output Resistance in Saturation

- **Q:** What effect will increased v_{DS} have on n -channel once pinch-off has occurred?
 - **A:** This voltage accelerates electrons as they reach the drain end, and sweep them across the “new” depletion region.
 - **A:** However, at the same time, the length of the n -channel will decrease.
 - Known as **channel length modulation**.
- This equation is valid when $v_{DS} \gg v_{ov}$, $v_{DS} \approx 0$

$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}^2 (1 + \lambda v_{DS}) \xrightarrow{v_{DS} \gg v_{ov}} i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}^2$$



Increasing v_{DS} beyond v_{DSsat} causes the channel pinch-off point to move slightly away from the drain, thus reducing the effective channel length by ΔL .



Large-Signal Equivalent Model of the n -channel MOSFET in saturation, incorporating the **finite output resistance** r_o . The output resistance models the linear dependence of i_D on v_{DS} and is given by the equation in this slide.

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MOS TRANSISTOR

Finite Output Resistance in Saturation

- How is r_o defined?
 - **Step #1:** Note that r_o is the 1/slope of i_D - v_{DS} characteristic.
 - **Step #2:** Define relationship between i_D and v_{DS} using (5.23).
 - **Step #3:** Take derivative of this function.
 - **Step #4:** Use above to define r_o .

$$r_o = \left[\frac{\partial i_D}{\partial v_{DS}} \right]_{v_{GS} = \text{const.}}^{-1}$$

$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}^2 (1 + \lambda v_{DS})$$

$$\begin{aligned} \frac{\partial i_D}{\partial v_{DS}} &= \frac{\partial}{\partial v_{DS}} \left(\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}^2 (1 + \lambda v_{DS}) \right) \\ &= \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}^2 \lambda \end{aligned}$$

$$r_o = \left[\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) v_{ov}^2 \lambda \right]_{v_{GS} = \text{const.}}^{-1}$$

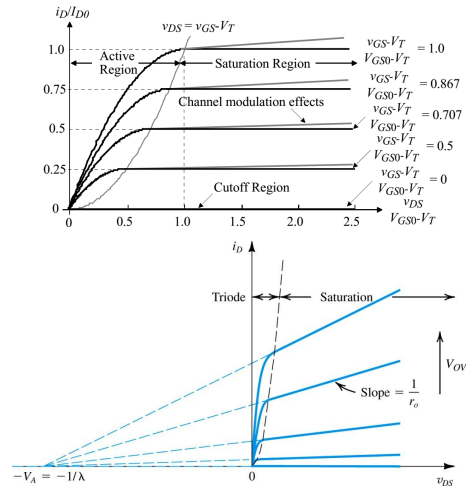
$$r_o = \frac{1}{\lambda i_D} = \frac{V_A}{i_D}$$

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MOS TRANSISTOR

Channel Length Modulation, λ

- What is channel length modulation λ ?
 - As the value of v_{DS} increases, the effective L decreases causing the current to increase.
 - A device parameter with the units of V^{-1} , the value of which depends on manufacturer's design and manufacturing process.
 - It can be much larger value for newer process.
- This figure demonstrates the effect of channel length modulation on v_{DS} - i_D curves
 - In short, we can draw a straight line between V_A and saturation to identify λ .
- We can neglect the effects of channel length modulation (assuming $\lambda = 0$) in DC circuits



Effect of v_{DS} on i_D in the saturation region. The MOSFET parameter V_A depends on the process technology and, for a given process, is proportional to the channel length L . 57

MOS TRANSISTOR

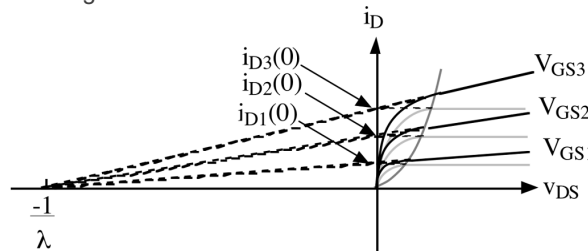
Channel Length Modulation Parameter, λ

- Assume the MOS is transistor is saturated

$$\therefore i_D = \frac{\mu C_{ox} W}{2L} (v_{GS} - V_T)^2 (1 + \lambda v_{DS})$$
- Define $i_D(0) = i_D$ when $v_{DS} = 0V$.

$$\therefore i_D(0) = \frac{\mu C_{ox} W}{2L} (v_{GS} - V_T)^2$$
- Now,

$$i_D = i_D(0)[1 + \lambda v_{DS}] = i_D(0) + \lambda i_D(0) v_{DS}$$
- Matching with $y = mx + b$ gives the value of λ

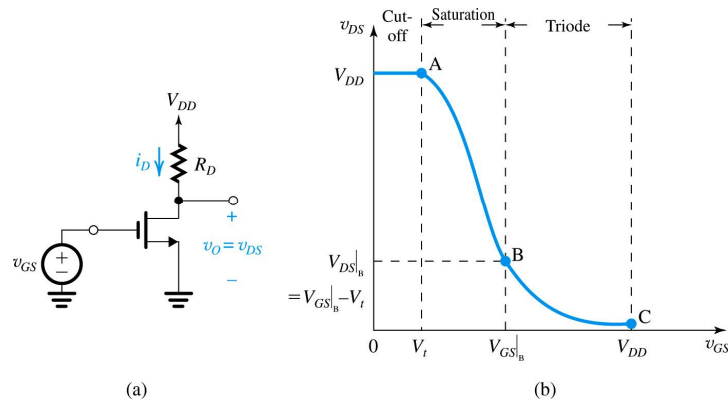


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MOS TRANSISTOR

Voltage Amplification

- Voltage controlled current source (VCCS) can serve as **transconductance amplifier**.
- Output voltage: $v_{DS} = V_{DD} - i_D R_D$

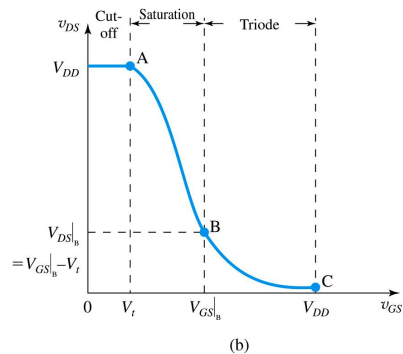


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MOS TRANSISTOR

Voltage Transfer Characteristics

- **Voltage transfer characteristics (VTC)** – plot of out voltage vs. input
- The voltage transfer characteristic (VTC) of the amplifier from previous slide



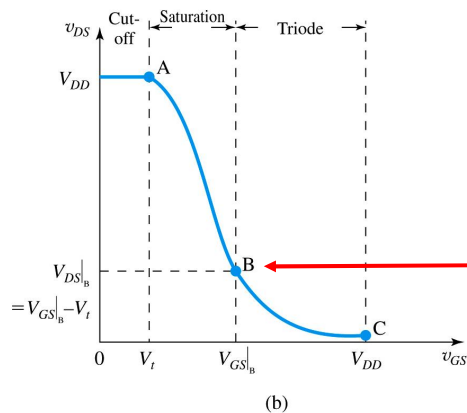
- **three regions** exist in VTC
 - $v_{GS} < V_t \rightarrow$ **cut off FET**
 - $v_{OV} = v_{GS} - V_t < 0$
 - $I_D = 0$
 - $v_{DS} < v_{OV}$
 - $v_{out} = v_{DD}$
 - $V_t < v_{GS} < v_{DS} + V_t \rightarrow$ **saturation**
 - $v_{OV} = v_{GS} - V_t > 0$
 - $I_D = \frac{1}{2} k_n (v_{GS} - V_t)^2$
 - $v_{DS} \gg v_{OV}$
 - $v_{out} = V_{DD} - I_D R_D$
 - $v_{DS} + V_t < v_{GS} < V_{DD} \rightarrow$ **triode/active**
 - $v_{OV} = v_{GS} - V_t > 0$
 - $I_D = k_n (v_{GS} - V_t - v_{DS}) v_{DS}$
 - $v_{DS} > v_{OV}$
 - $v_{out} = V_{DD} - I_D R_D$

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MOS TRANSISTOR

Voltage Transfer Characteristics

- The voltage transfer characteristic (VTC) of the amplifier from previous slide



$$v_{DS} = V_{DD} - \underbrace{\left[\frac{1}{2} k_n (v_{GS} - V_t)^2 \right]}_{I_D} R_D$$

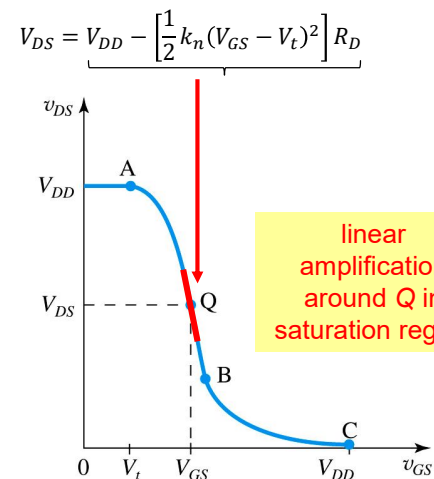
$$v_{GS}|_B = V_t + \frac{\sqrt{2k_n R_D V_{DD}} + 1 - 1}{k_n R_D}$$

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MOS TRANSISTOR

Voltage Transfer Characteristics

- How can we linearize VTC?
 - Appropriate biasing technique
 - DC voltage v_{GS} is selected to obtain operation at point Q on segment AB
- Biasing the MOSFET to obtain Linear Amplification
- bias point/dc operating pt. (Q)** – point of linearization for MOSFET (also known as **quiescent point.**)
- Because VTC is linear near Q, we may perform linear amplification of signal $\ll Q$.



Biasing the MOSFET amplifier at point Q located on segment AB of VTC

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MOS TRANSISTOR

Biasing the MOSFET to Obtain Linear Amplification

- How is **linear gain** achieved?

- Step #1:** Bias MOSFET with DC voltage V_{GS} as defined by

$$v_{DS} = V_{DD} - \left[\frac{1}{2} k_n (v_{GS} - V_t)^2 \right] R_D$$

- Step #2:** Superimpose amplifier input ($v_{gs}(t)$) upon V_{GS} .

$$v_{GS}(t) = V_{GS} + v_{gs}(t)$$

- Step #3:** Resultant v_{ds} should be linearly proportional to small-signal component v_{gs} .

$$v_{ds}(t) \propto v_{gs}(t)$$

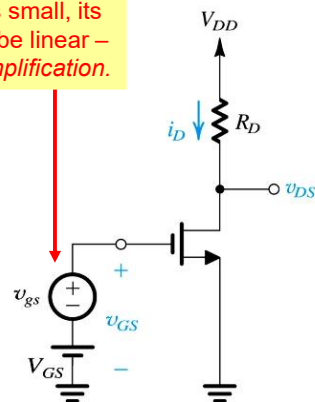
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MOS TRANSISTOR

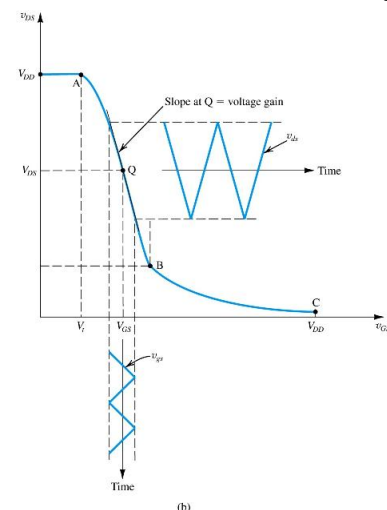
Biasing the MOSFET to obtain Linear Amplification

- How is **linear gain** achieved?

As long as $v_{GS}(t)$ is small, its effect on $v_{DS}(t)$ will be linear – facilitating linear amplification.



The MOSFET amplifier with a small time-varying signal $v_{gs}(t)$ superimposed on the dc bias voltage V_{GS} . The MOSFET operates on a short almost-linear segment of the VTC around the bias point Q and provides an output voltage $v_{ds} = A_v v_{gs}$



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MOS TRANSISTOR

Biasing the MOSFET to obtain Linear Amplification

- How is **linear gain** achieved?
 - Step #4:** Note if v_{gs} is small, output v_{ds} will be nearly **linearly proportional** to it. Slope will be constant.
 - Gain is **negative**.
 - Gain is **proportional** to:
 - load resistance (R_D)
 - transistor conductance parameter (k_n)
 - overdrive voltage (v_{ov})
 - Gain is **ratio** of:
 - voltage drop across R_D
 - half of over voltage

$$A_v = \left. \frac{dv_{DS}}{dv_{GS}} \right|_{v_{GS} \approx V_{GS}} \quad (\text{Means that } v_{GS} \text{ is small})$$

$$A_v = \left. \frac{d \left(V_{DD} - \left[\frac{1}{2} k_n (v_{GS} - V_t)^2 \right] R_D \right)}{dv_{GS}} \right|_{v_{GS} \approx V_{GS}}$$

$$A_v = -k_n (V_{GS} - V_t) R_D$$

$$A_v = -k_n V_{ov} R_D \quad i_D = \frac{1}{2} k_n V_{ov}^2$$

$$A_v = - \left(\frac{I_D R_D}{V_{ov}/2} \right)$$

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MOS TRANSISTOR

Biasing the MOSFET to obtain Linear Amplification

- How does **this equation** $A_v = - \left(\frac{I_D R_D}{V_{ov}/2} \right)$ **relate** to physical devices?
 - For modern CMOS technology, V_{ov} is **usually no less** than $0.2V$.
 - This means that **max achievable gain** is approximately $10V_{DD}$.

$$\max(A_v) = - \left(\frac{\overbrace{\max(I_D) R_D}^{V_{DD}}}{\underbrace{V_{ov}/2}_{0.1V}} \right) = -10V_{DD}$$

This does not mean that output may be 10x supply (V_{DD}).

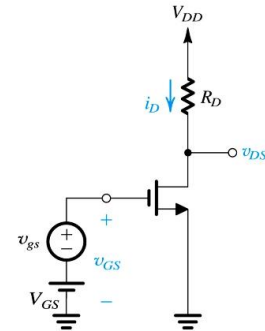
- For example, $0.13\mu m$ CMOS technology with $V_{DD} = 1.3V$ yields maximum gain of $13V/V$.

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MOS TRANSISTOR

Exercise 3:

- Consider the amplifier circuit shown in the Figure. The transistor is specified to have $V_t = 0.4\text{V}$, $k_n = 0.4\text{mA/V}^2$, $W/L = 10$, and $I = 0$. Also, let $V_{DD} = 1.8\text{V}$, $R_D = 17.5\text{k}\Omega$, and $V_{GS} = 0.6\text{V}$.
- a) For $v_{gs} = 0$ (and hence $v_{ds} = 0$), find V_{OV} , I_D , V_{DS} , and A_v .
- b) What is the maximum symmetrical signal swing allowed at the drain? Hence, find the maximum allowable amplitude of a sinusoidal v_{gs} .



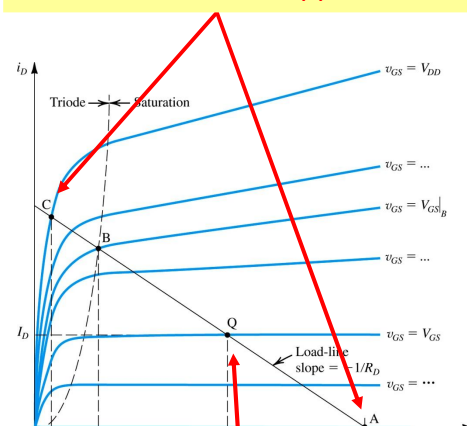
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MOS TRANSISTOR

Determining the VTC via Graphical Analysis

- **point A** – where $v_{GS} = V_t$
- **point Q** – where MOSFET may be biased for amplifier operation
 - $v_{GS} = V_{GS}$, $v_{DS} = V_{DS}$
- **point B** – where MOSFET leaves saturation / enters triode
- **point C** – where MOSFET is deep in triode region and $v_{GS} = V_{DD}$

Points A (open) and C (closed) are suitable for switch applications



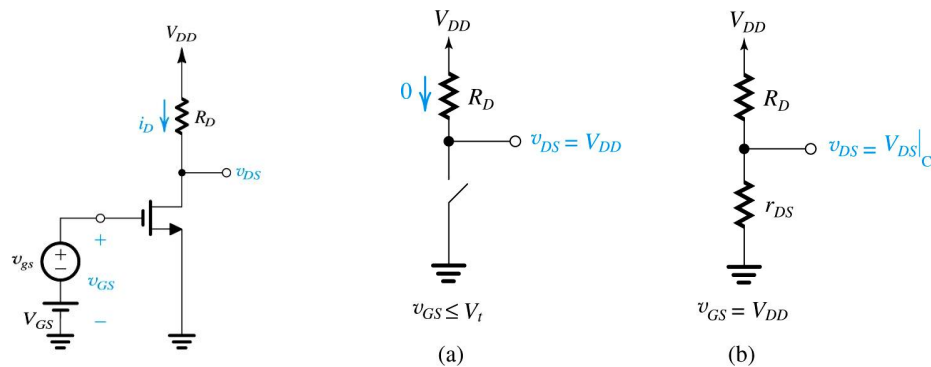
Point Q is suitable for amplifier applications

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MOS TRANSISTOR

Determining the VTC via Graphical Analysis

- Operation of the MOSFET Amplifier as a switch: **(a)** Open, corresponding to point A; **(b)** Closed, corresponding to point C. The closure resistance is approximately equal to r_{DS} because V_{DS} is usually very small.

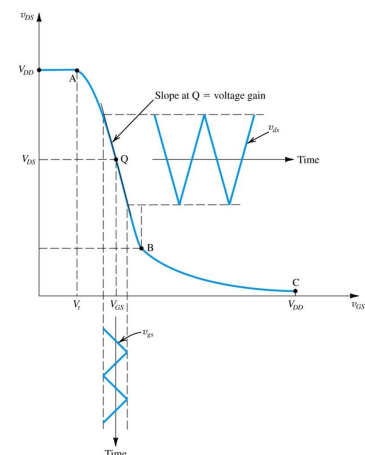


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MOS TRANSISTOR

Locating the Bias Point Q

- bias point (Q)** – is determined by value of v_{GS} and load resistance R_D .
- Two considerations** in deciding Q:
 - Required **gain**.
 - Allowable signal **swing** at output.
- How is **Q for VTC defined** (assuming R_D is fixed)?
 - As point Q approaches **B**:
 - gain increases
 - maximum v_{GS} swing decreases

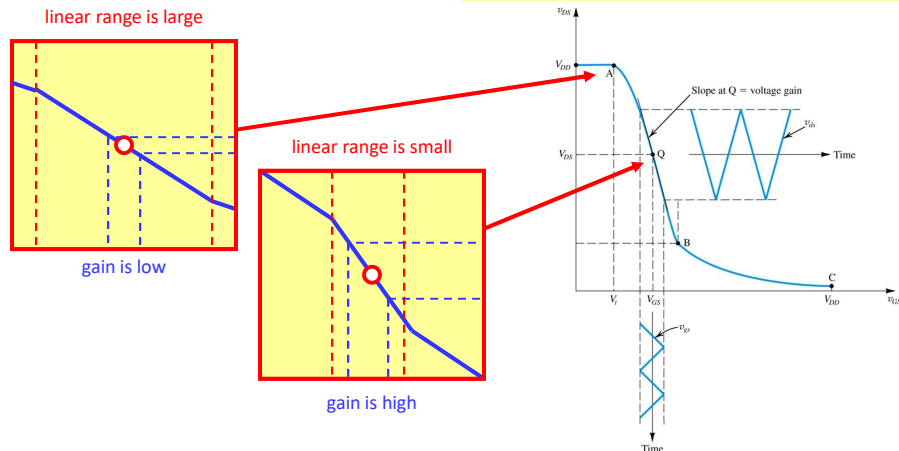


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MOS TRANSISTOR

Locating the Bias Point Q

Note that a trade-off between gain and linear range exists.



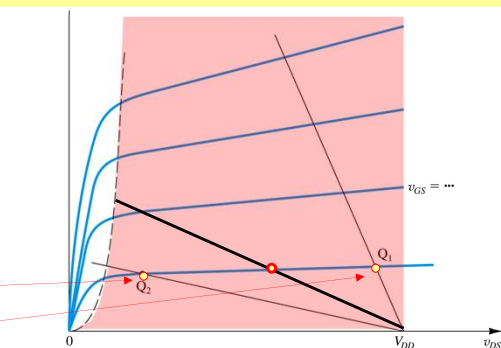
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MOS TRANSISTOR

Locating the Bias Point Q

The objective is to prevent v_{DS} from "clipping" or entering triode region

- To define **load resistance R_D** , one should refer to the i_D - v_{DS} plane.
- Two examples of R_D are shown to right for illustration:
 - Q2**: too close to triode
 - not enough **legroom**
 - Q1**: too close to V_{DD}
 - not enough **headroom**
- Ideally, we want to be somewhere in the **middle**.



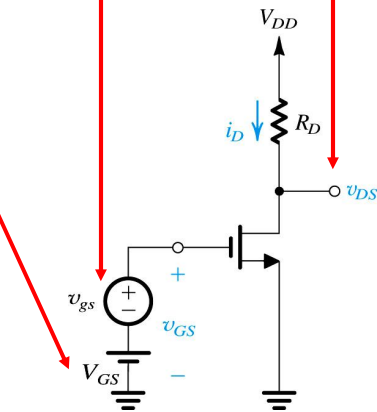
Two load lines and corresponding bias points. Bias point Q_1 does not leave sufficient room for positive signal swing at the drain (too close to V_{DD}). Bias point Q_2 is too close to the boundary of the triode region and might not allow for sufficient negative signal swing.

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MOS TRANSISTOR

Small-Signal Operation and Models

- Previously it was stated that **linear amplification may be obtained from MOSFET** via...
 - Operation in **saturation** region
 - Utilization of **small-input**
- This section will explore **small-signal operation** in detail
 - Note the **conceptual amplifier** circuit to right



Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

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MOS TRANSISTOR

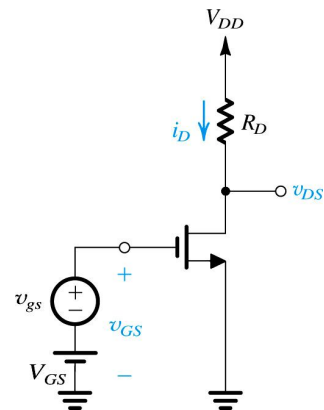
The DC Bias Point

- How is **dc bias current** I_D defined?

$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2 = \frac{1}{2} k_n V_{ov}^2$$

Only applies in saturation
where $V_{DS} > V_{ov}$

$$V_{DS} = V_{DD} - R_D I_D$$



Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

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MOS TRANSISTOR

The DC Bias Point

- What is **effect of v_{gs}** on i_D ?
 - step #1:** Define v_{GS} as $v_{GS} = V_{GS} + v_{GS}$
 - step #2:** Define i_D , **separate terms** as function of V_{GS} and v_{gs}

$$i_D = \frac{1}{2} k_n V_{ov}^2 = \frac{1}{2} k_n (v_{GS} - V_t)^2 = \frac{1}{2} k_n (V_{GS} + v_{GS} - V_t)^2$$

$$i_D = \frac{1}{2} k_n [(V_{GS} - V_t)^2 + 2(V_{GS} - V_t)v_{GS} + v_{GS}^2]$$

Expand the squared term via $V_{GS} - V_t$ and v_{GS}

$$i_D = \frac{1}{2} k_n \underbrace{(V_{GS} - V_t)^2 + k_n (V_{GS} - V_t)v_{GS} + \frac{1}{2} k_n v_{GS}^2}_{\text{Simplify}}$$

Note that this differs from previous analyses - because of attempt to isolate the effect of v_{gs} from V_{GS} .

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MOS TRANSISTOR

The DC Bias Point

- What is **effect of v_{gs}** on i_D ?
 - step #3: Classify** terms.
 - dc bias** current (I_D).
 - linear gain** – is desirable.
 - nonlinear distortion** – is undesirable, because rep. distortion.

Note that to minimize nonlinear distortion, v_{gs} should be kept small.

$$\frac{1}{2} k_n v_{gs}^2 \ll k_n (V_{GS} - V_t) v_{gs}$$

$$v_{gs} \ll 2(V_{GS} - V_t)$$

$$v_{gs} \ll 2V_{OV}$$

$$i_D = \underbrace{\frac{1}{2} k_n (V_{GS} - V_t)^2}_{\text{DC bias current}} + \underbrace{k_n (V_{GS} - V_t) v_{GS}}_{\text{Linear gain term}} + \underbrace{\frac{1}{2} k_n v_{GS}^2}_{\text{Nonlinear distortion term}}$$

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MOS TRANSISTOR

The DC Bias Point

- What is effect of v_{gs} on i_D ?
 - step #4:** Adapt the following equation for small-signal condition.
 - If $v_{gs} \ll 2V_{OV}$, neglect distortion.

$$i_D = \underbrace{\frac{1}{2}k_n(V_{GS} - V_t)^2}_{\text{DC bias current } (I_D)} + \underbrace{k_n(V_{GS} - V_t)v_{GS}}_{\text{Linear gain term}} + \underbrace{\frac{1}{2}k_nv_{GS}^2}_{\text{Nonlinear distortion term}}$$

- MOSFET Transconductance

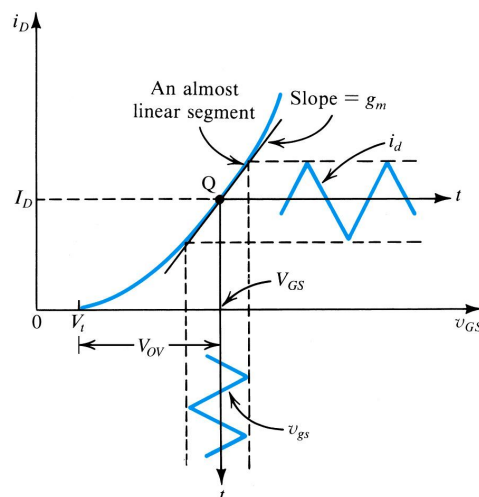
$$g_m = \frac{v_{GS}}{i_D} = k_n(V_{GS} - V_t)$$

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MOS TRANSISTOR

- Small-signal operation of the MOSFET amplifier

$$g_m \equiv \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS} = V_{GS}}$$



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MOS TRANSISTOR

The gain voltage

- How is **voltage gain** (A_v) defined?

- step #1:** Define v_{DS} for circuit of the Figure using KVL.

$$v_{DS} = V_{DD} - R_D i_D = V_{DD} - R_D (I_D + i_d)$$

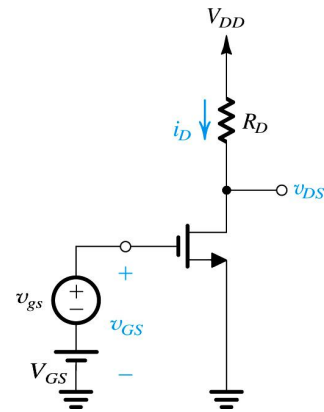
action: apply small-signal condition

$$v_{DS} = \underbrace{V_{DD} - R_D I_D}_{\text{dc component (} V_{DS} \text{)}} - R_D i_d = V_{DS} - \underbrace{R_D i_d}_{v_{ds}}$$

action: regroup terms action: simplify

- step #2:** Isolate v_{ds} component of v_{DS} : $v_{DS} = -R_D i_D = -R_D (g_m v_{GS})$
- step #3:** Solve for **gain** (A_v).

$$A_v = \frac{v_{DS}}{v_{GS}} = -g_m R_D$$



Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

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MOS TRANSISTOR

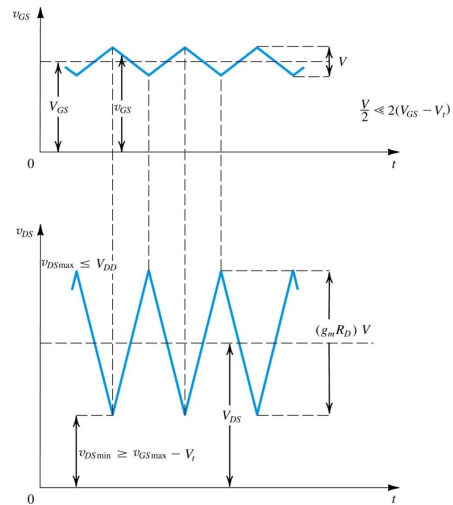
The gain voltage

- Output signal is **shifted** from input by 180°.
- Input signal $v_{gs} \ll 2(V_{GS} - V_t)$.
- Operation should remain in MOSFET **saturation** region
 - $V_{DS} > V_{GS} - V_t$ (**legroom**)
 - $V_{DS} < V_{DD}$ (**headroom**)

Input signals are generally time varying signals with positive and negative cycles.

The range which an amplifier provides for amplifying the positive cycle of the input signal is termed as the '**headroom**' provided by the amplifier before it will leave the region of operation and stop working as an amplifier.

The range which an amplifier provides for amplifying the negative cycle of the input signal is termed as '**legroom**' of the amplifier.



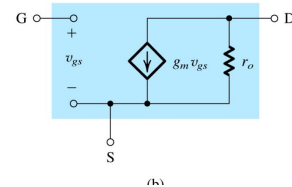
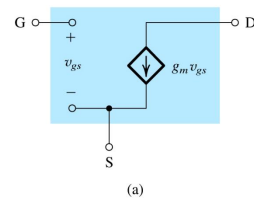
Total instantaneous voltage v_{GS} and v_{DS} for the circuit.

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MOS TRANSISTOR

Small-signal Equivalent Models

- FET behaves as VCCS.
 - Accepts v_{gs} between gate and source
 - Provides current (i_D) at drain
 - Input resistance is high
 - b/c gate terminal draws $i_G = 0$
 - Output resistance is high



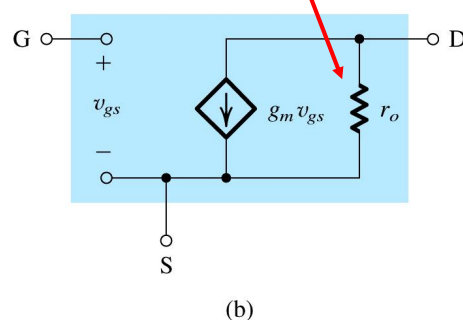
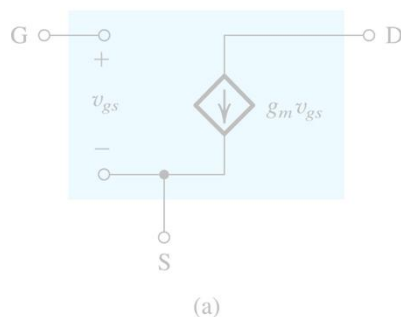
Small-signal models for the MOSFET: **(a)** neglecting the dependence of i_D on v_{DS} in saturation (the channel-length modulation effect) and **(b)** including the effect of channel length modulation

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MOS TRANSISTOR

Small-signal Equivalent Models

Note that this resistor (r_o) takes on value $10k\Omega$ to $1M\Omega$ and represents channel-length modulation.



Small-signal models for the MOSFET: **(a)** neglecting the dependence of i_D on v_{DS} in saturation (the channel-length modulation effect) and **(b)** including the effect of channel length modulation

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MOS TRANSISTOR

The Transconductance g_m

- g_m is proportional to μ_n , C_{ox} , ratio W/L , DC component V_{OV} .
- MOSFET with short / wide channel provides maximum gain.
- Gain may be increased via V_{GS} , but not without reducing allowable swing of v_{gs} .

$$g_m = \frac{i_d}{v_{gs}} = k_n (V_{GS} - V_t)$$

action: make some substitutions

$$g_m = k'_n \frac{W}{L} (V_{GS} - V_t)$$

action: simplify

$$g_m = k'_n \frac{W}{L} V_{OV}$$

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MOS TRANSISTOR

The Transconductance g_m

- g_m is proportional to square root of dc bias current (I_D)
- For given I_D , g_m is proportional to $(W/L)^{1/2}$

$$I_D = \frac{1}{2} k'_n \frac{W}{L} V_{OV}^2$$

action: solve for V_{OV}

$$V_{OV} = \frac{\sqrt{2I_D}}{\sqrt{k'_n} \sqrt{W/L}}$$

$$g_m = k'_n \frac{W}{L} V_{OV}$$

action: substitute for V_{OV} as defined above

$$g_m = k'_n \frac{W}{L} \frac{\sqrt{2I_D}}{\sqrt{k'_n} \sqrt{W/L}}$$

action: simplify

$$g_m = \sqrt{2k'_n} \sqrt{W/L} \sqrt{I_D}$$

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MOS TRANSISTOR

The Transconductance g_m

- The figure illustrates the relationship defined in the last equation.

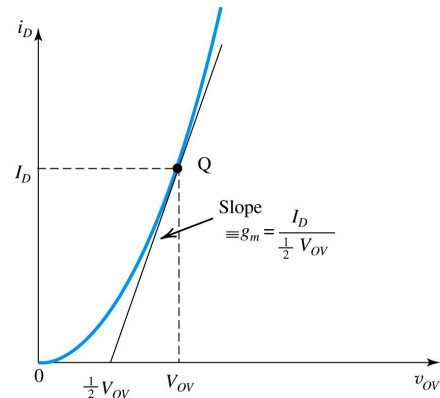
$$g_m = k'_n \frac{W}{L} V_{OV}$$

action: replace $k'_n \frac{W}{L}$

$$g_m = \left(\frac{2I_D}{(V_{GS} - V_t)^2} \right) V_{OV}$$

action: simplify

$$g_m = \frac{2I_D}{V_{GS} - V_t} = \frac{2I_D}{V_{OV}}$$



The slope of the tangent at the bias point Q intersects the v_{OV} axis at $1/2 V_{OV}$. Thus $g_m = I_D / (1/2 V_{OV})$.

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MOS TRANSISTOR

The Transconductance g_m

- In summary, there are **three relationships** for determining g_m :

$$g_m = k'_n \frac{W}{L} V_{OV}$$

$$g_m = \sqrt{2k'_n} \sqrt{W/L} \sqrt{I_D}$$

$$g_m = \frac{2I_D}{V_{OV}}$$

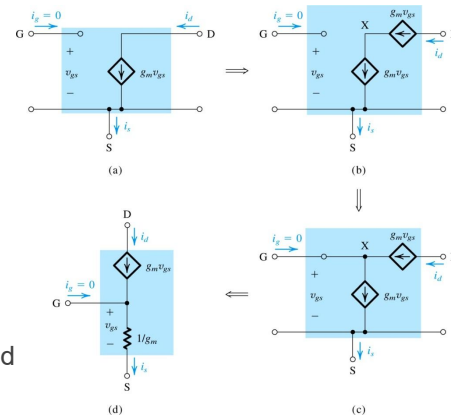
- These relationships are dependent on **three design parameters**:
 - W/L , V_{OV} , I_D

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MOS TRANSISTOR

The T Equivalent-Circuit Model

- Through circuit transformation, it is possible to develop **alternative circuit models**
- T-Equivalent-Ckt Model** is shown to right.
- Review equivalent circuit in the subject “Signals and Systems” or “Circuit Analysis and Systems”



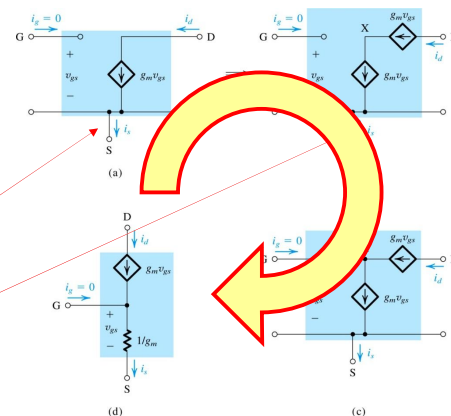
Development of the T equivalent-circuit model for the MOSFET. For simplicity, r_o has been omitted; however, it may be added between D and S in the T model of (d).

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MOS TRANSISTOR

The T Equivalent-Circuit Model

- How is this model developed?
 - step #1:** Begin with **small signal model** (assume $R_o=0$).
 - step #2:** Place second **current source in series** with the first.
 - Has no effect on circuit operation.



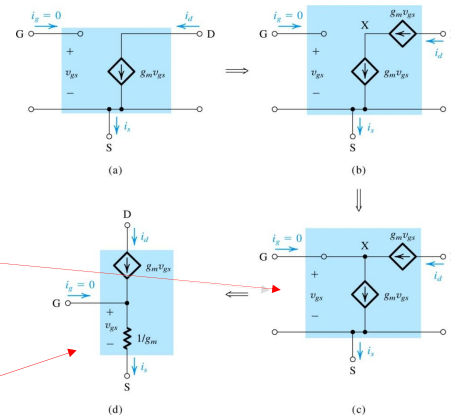
Development of the T equivalent-circuit model for the MOSFET. For simplicity, r_o has been omitted; however, it may be added between D and S in the T model of (d).

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MOS TRANSISTOR

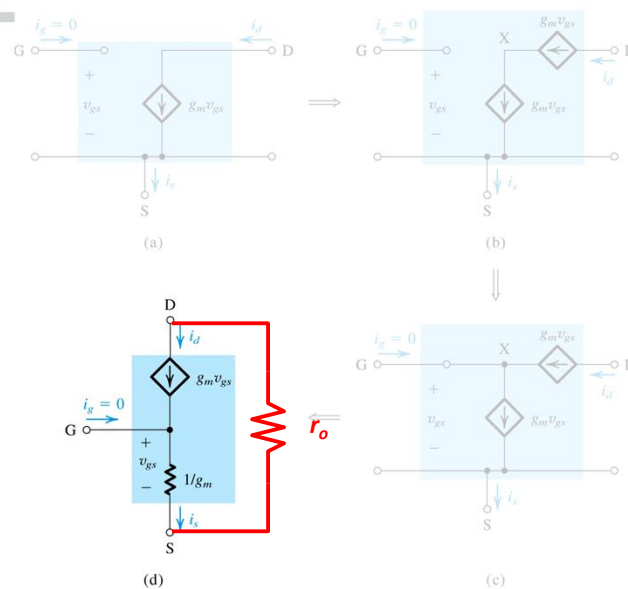
The T Equivalent-Circuit Model

- How is this model developed?
 - step #3:** Create **new node X**, which connects gate and drain terminals
 - b/c the two current sources are equal, $i_g = 0$
 - step #4:** replace initial current source with **equivalent resistance**.
 - $i_{DS} = g_m v_{gs} = v_{gs} / R_{gs}$



Development of the T equivalent-circuit model for the MOSFET. For simplicity, r_o has been omitted; however, it may be added between D and S in the T model of **(d)**.

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Development of the T equivalent-circuit model for the MOSFET. For simplicity, r_o has been omitted; however, it may be added.

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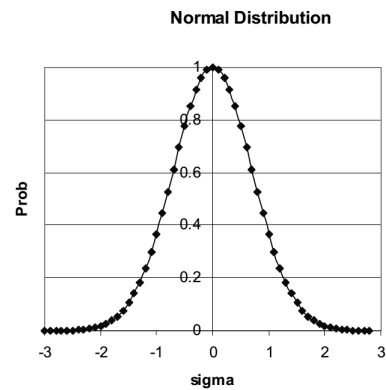
Table 5.3 Small-Signal Equivalent-Circuit Models for the MOSFET	
Small-Signal Parameters	
NMOS transistors	
■ Transconductance:	$g_m = \mu_n C_{ox} \frac{W}{L} V_{OV} = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = \frac{2I_D}{V_{OV}}$
■ Output resistance:	$r_o = V_A / I_D = 1 / \lambda I_D$
PMOS transistors	
Same formulas as for NMOS <i>except</i> using $ V_{OV} $, $ V_A $, and replacing μ_n with μ_p .	
Small-Signal Equivalent Circuit Models	
The figure shows three small-signal equivalent circuit models for a MOSFET. The first model is the Hybrid-π model, showing a dependent current source $g_m v_{gs}$ in parallel with the output resistance r_o, connected between the drain (D) and source (S) terminals, with the gate (G) terminal as the input. The second model is a T model, showing a dependent current source $g_m v_{gs}$ in series with the output resistance r_o, connected between the drain (D) and source (S) terminals, with the gate (G) terminal as the input. The third model is another T model, showing a dependent current source $g_m v_{gs}$ in series with the output resistance r_o, connected between the drain (D) and source (S) terminals, with the gate (G) terminal as the input.	

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DEPENDENCE OF THE LARGE SIGNAL MODEL ON PVT

Process Variations

- We have talked about transistors as if all transistors were the same
- But, in fact that they are not exactly alike
- Parameters of a fabrication run are generally normally distributed: mean, standard deviation (in Gauss distribution)
- Parameters can vary depending on Process, Voltage, and Temperature (PVT)
 - Process: Threshold voltage (NMOS, PMOS), Metal width, Oxide thickness, Etching, Doping, etc
 - Voltage: Operating voltage (can be high, typical, low)/Power supply
 - Temperature (can be hot, typical, cold)
- Transistors (NMOS, PMOS) can be slow (S), typical (T), fast (F)

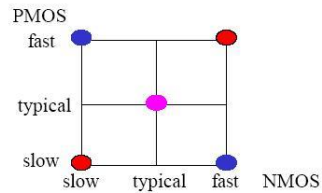


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DEPENDENCE OF THE LARGE SIGNAL MODEL ON PVT

Process Corners

- In this naming convention, three corners exist: typical, fast and slow.
 - Fast corners exhibit carrier mobilities that are higher than normal.
 - Slow corners exhibit carrier mobilities that are lower than normal.
- There are five possible corners: typical-typical (TT), fast-fast (FF), slow-slow (SS), fast-slow (FS), and slow-fast (SF).
 - The typical corner is stand for the typical case
 - The SS corner is stand for slow NMOS and slow PMOS case
 - The FF corner is fast NMOS and fast PMOS case
 - The FS corner is fast NMOS and slow PMOS case
 - The SF corner is slow NMOS and fast PMOS case
- The first three corners (TT, FF, SS) are called even corners, because both types of devices are affected evenly, and generally do not adversely affect the logical correctness of the circuit. The resulting devices can function at slower or faster clock frequencies. The last two corners (FS, SF) are called "skewed" corners, and are cause for concern.

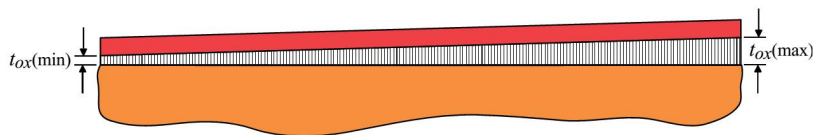


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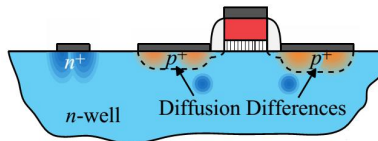
DEPENDENCE OF THE LARGE SIGNAL MODEL ON PROCESS

How Does Technology Vary?

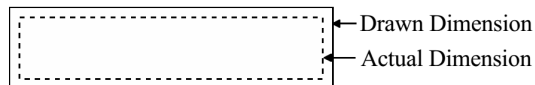
- Thickness variations in layers (dielectrics and metal)



- Doping variations



- Process biases – differences between the drawn and actual dimensions due to process (etching, lateral diffusion, etc.)



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DEPENDENCE OF THE LARGE SIGNAL MODEL ON PROCESS

Large Signal Model Dependence on Process Variations

1. Threshold voltage

$$V_T = V_{T0} + \gamma(\sqrt{|-2\phi_F + v_{SB}|} - \sqrt{|-2\phi_F|})$$

$$\text{where } V_{T0} = \phi_{MS} - 2\phi_F - \frac{Q_{b0}}{C_{ox}} - \frac{Q_{SS}}{C_{ox}} \quad \gamma = \frac{\sqrt{2q\epsilon_{Si}N_A}}{C_{ox}}$$

- If $V_{BS} = 0$, then V_T is dependent on doping and oxide thickness because

$$\phi_F = \frac{kT}{q} \ln\left(\frac{N_{SUB}}{n_i}\right) \quad C_{ox} \propto \frac{1}{t_{ox}}$$

(Recall that the threshold is also determined by the threshold implant during processing)

2. Transconductance parameter

$$K' = \mu_o C_{ox} \propto \frac{1}{t_{ox}}$$

- For short channel devices, the mobility is degraded as given by

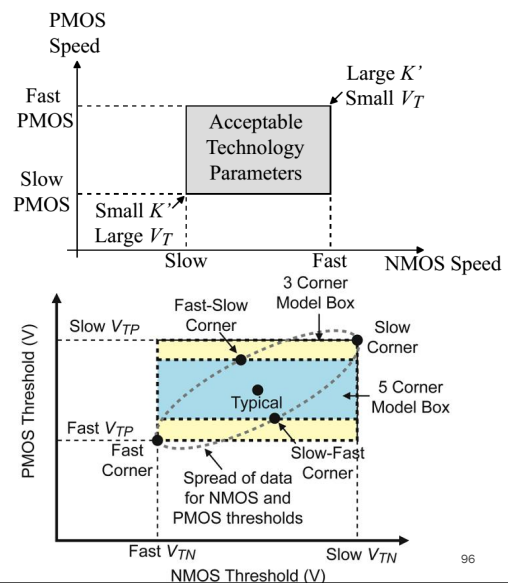
$$\mu_{eff} = \frac{\mu_o}{1 + \theta(V_{GS} - V_T)} \quad \theta \approx \frac{2 \times 10^{-9} \text{ m/V}}{t_{ox}}$$

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DEPENDENCE OF THE LARGE SIGNAL MODEL ON PROCESS

Process Variations "Corners"

- For strong inversion operation, the primary influence is the oxide thickness, t_{ox} . We see that K' will tend to increase with decreasing oxide thickness whereas V_T tends to decrease.
- If the "speed" of a transistor is increased by increasing K' and decreasing V_T , then the variation of technology can be expressed on a two-dimensional graph resulting in a rectangular area of "acceptable" process limitation.
- Three corner versus five corner models



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DEPENDENCE OF THE LARGE SIGNAL MODEL ON VOLTAGE

What is Voltage Variation?

- Voltage variation is the influence of power supply voltage on the component.
(There is also power supply influence on the circuit called power supply rejection ratio, PSRR. We will deal with this in a later lecture.)
- Power supply variation comes from:
 1. Influence of depletion region widths on components.
 2. Nonlinearity (e.g., velocity saturation)
 3. Breakdown voltage
- Note: Because the large-signal model for the MOSFET includes all the influences of voltage on the transistor, we will focus on passive components except for breakdown.

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DEPENDENCE OF THE LARGE SIGNAL MODEL ON VOLTAGE

Models for Voltage Dependence of a Component

1. i^{th} -order Voltage Coefficients
 - In general, a variable $y = f(v)$ which is a function of voltage, v , can be expressed as a Taylor series,

$$y(v = V_0) \approx y(V_0) + a_1(v - V_0) + a_2(v - V_0)^2 + a_3(v - V_0)^3 + \dots$$
 where the coefficients, a_i , are defined as,

$$a_1 = \left. \frac{df(v)}{dv} \right|_{v=V_0}, a_2 = \frac{1}{2} \left. \frac{d^2f(v)}{dv^2} \right|_{v=V_0}, \dots$$
 - The coefficients, a_i , are called the first-order, second-order, voltage coefficients.
2. Fractional Voltage Coefficient or Voltage Coefficient
 - Generally, only the first-order coefficients are of interest.
 - In the characterization of temperature dependence, it is common practice to use a term called fractional voltage coefficient, VC_F , which is defined as,

$$VC_F(v=V_0) = \frac{1}{f(v=V_0)} \left. \frac{df(v)}{dv} \right|_{v=V_0} \text{ parts per million/V (ppm/V)}$$
 - or more simply,

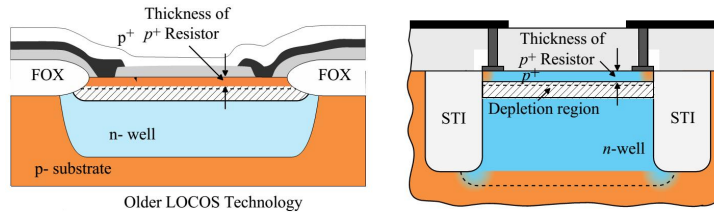
$$VC_F = \frac{1}{f(v)} \frac{df(v)}{dv} \text{ parts per million/V (ppm/V)}$$

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DEPENDENCE OF THE LARGE SIGNAL MODEL ON VOLTAGE

Influence of Voltage on a Diffused Resistor – Depletion Region

- Influence of the depletion region on the p⁺ resistor:



- As the voltage at the terminals of the resistor become smaller than the n-well potential, the depletion region will widen causing the thickness of the resistor to decrease.

$$R = \frac{\rho L}{tW} \propto \sqrt{V_R}$$

where V_R is the reverse bias voltage from the resistor to the well.

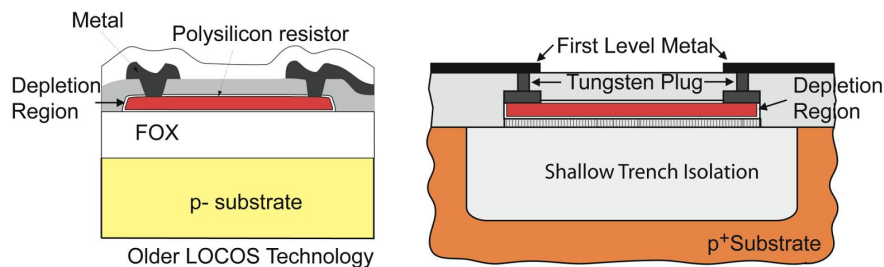
- This effect is worse for well resistors because the doping concentration of the resistor is smaller.
 - Voltage coefficient for diffused resistors $\approx 200\text{--}800$ ppm/V
 - Voltage coefficient for well resistors ≈ 8000 ppm/V

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DEPENDENCE OF THE LARGE SIGNAL MODEL ON VOLTAGE

Voltage Coefficient of Polysilicon Resistors

- Why should polysilicon resistors be sensitive to voltage?
- There is a small depletion region between the polysilicon and its surrounding material that has a very small dependence on the voltage between the polysilicon and the surrounding material.

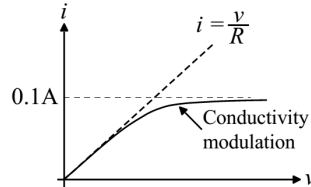


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DEPENDENCE OF THE LARGE SIGNAL MODEL ON VOLTAGE

Voltage Nonlinearity and Breakdown Voltage

- Conductivity modulation: As the current in a resistor increases, the conductivity becomes modulated and the resistance increases.

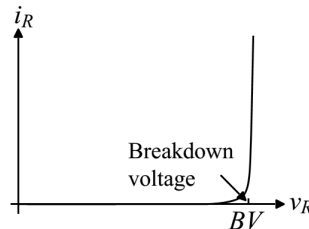


- Example of a n-well resistor:

- As the reverse bias voltage across a pn junction becomes large, at some point, called the breakdown voltage, the current will rapidly increase. Both transistors, diodes and depletion capacitors experience this breakdown.

- Model for current multiplication factor:

$$i_R = M \cdot I_R \text{ where } M = \frac{1}{1 - \left(\frac{v_R}{BV}\right)^n}$$



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DEPENDENCE OF THE LARGE SIGNAL MODEL ON TEMPERATURE

Temperature Dependence of the MOSFET

- Transconductance parameter:

$$K'(T) = K'(T_0) (T/T_0)^{-1.5}$$

- Threshold Voltage:

$$V_T(T) = V_T(T_0) + \alpha(T - T_0) + \dots$$

- Typically, $\alpha_{NMOS} = -2\text{mV}/^\circ\text{C}$ to $-3\text{mV}/^\circ\text{C}$ from 200°K to 400°K (PMOS has a + sign)
- Example:** Find the value of I_D for a NMOS transistor at 27°C and 100°C if $V_{GS} = 2\text{V}$ and $W/L = 5\mu\text{m}/1\mu\text{m}$ if $K'(T_0) = 110\mu\text{A}/\text{V}^2$ and $V_T(T_0) = 0.7\text{V}$ and $T_0 = 27^\circ\text{C}$ and $\alpha_{NMOS} = -2\text{mV}/^\circ\text{C}$.

- Solution:**

- At room temperature, the value of drain current is,

$$I_D(27^\circ\text{C}) = \frac{110\mu\text{A}/\text{V}^2 \cdot 5\mu\text{m}}{2 \cdot 1\mu\text{m}} (2 - 0.7)^2 = 465\mu\text{A}$$

- At $T = 100^\circ\text{C}$ (373K), $K'(100^\circ\text{C}) = K'(27^\circ\text{C}) (373/300)^{-1.5} = 110\mu\text{A}/\text{V}^2 \cdot 0.72 = 79.3\mu\text{A}/\text{V}^2$

$$I_D(100^\circ\text{C}) = \frac{79.3\mu\text{A}/\text{V}^2 \cdot 5\mu\text{m}}{2 \cdot 1\mu\text{m}} (2 - 0.554)^2 = 415\mu\text{A} \quad (\text{Repeat with } V_{GS} = 2.0855\text{V})$$

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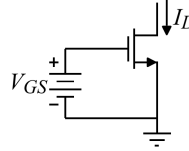
DEPENDENCE OF THE LARGE SIGNAL MODEL ON TEMPERATURE

Zero Temperature Coefficient (ZTC) Point for MOSFETs

- For a given value of gate-source voltage, the drain current of the MOSFET will be independent of temperature. Consider the following circuit:

- Assume that the transistor is saturated and that:

$$\mu = \mu_0 \left(\frac{T}{T_0} \right)^{-1.5} \quad \text{and} \quad V_T(T) = V_T(T_0) + \alpha(T - T_0)$$



$$\therefore I_D(T) = \frac{\mu_0 C_{ox} W}{2L} \left(\frac{T}{T_0} \right)^{-1.5} [V_{GS} - V_{T0} - \alpha(T - T_0)]^2$$

$$\frac{dI_D}{dT} = \frac{-3\mu_0 C_{ox} W}{4LT_0} \left(\frac{T}{T_0} \right)^{-2.5} [V_{GS} - V_{T0} - \alpha(T - T_0)]^2 - \alpha \frac{\mu_0 C_{ox} W}{L} \left(\frac{T}{T_0} \right)^{-1.5} [V_{GS} - V_{T0} - \alpha(T - T_0)] = 0$$

$$\therefore V_{GS} - V_{T0} - \alpha(T - T_0) = \frac{-4T\alpha}{3} \Rightarrow V_{GS}(ZTC) = V_{T0} - \alpha T_0 - \frac{\alpha T}{3}$$

- Let $K' = 20 \mu\text{A/V}^2$, $W/L = 1.5/4$, $\alpha = -0.0016 \text{V/}^\circ\text{C}$ and $V_{T0} = 0.46 \text{V}$.
- At $T = 27^\circ\text{C}$ (300°K), $V_{GS}(ZTC) = 0.46 - (-0.0016)(300^\circ\text{K}) - (0.333)(-0.0016)(300^\circ\text{K}) = 1.10 \text{V}$
- At $T = 27^\circ\text{C}$ (300°K), $I_D = (20 \mu\text{A/V}^2)(1.5/4)(1.10 - 0.46)^2 = 3.07 \mu\text{A}$

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DEPENDENCE OF THE LARGE SIGNAL MODEL ON TEMPERATURE

Temperature Modeling of the PN Junction

- PN Junctions (Reverse-biased only):

$$-i_D \cong I_s = qA \left[\frac{D_{ppn0}}{L_p} + \frac{D_{nnp0}}{L_n} \right] \cong \frac{qAD}{L} \frac{n_i^2}{N} = KT^3 \exp \left(\frac{-V_{Go}}{V_t} \right)$$

- Differentiating with respect to temperature gives,

$$\frac{dI_s}{dT} = \frac{3KT^3}{T} \exp \left(\frac{-V_{Go}}{V_t} \right) + \frac{qKT^3 V_{Go}}{KT^2} \exp \left(\frac{-V_{Go}}{V_t} \right) = \frac{3I_s}{T} + \frac{I_s V_{Go}}{T V_t}$$

$$TC_F = \frac{dI_s}{I_s dT} = \frac{3}{T} + \frac{1}{T} \frac{V_{Go}}{V_t}$$

- Example:** Assume that the temperature is 300°K (room temperature) and calculate the reverse diode current change and the TC_F for a 5°K increase.
- Solution:** The TC_F can be calculated from the above expression as $TC_F = 0.01 + 0.155 = 0.165$.
- Since the TC_F is changed per degree, the reverse current will increase by a factor of 1.165 for every degree K (or $^\circ\text{C}$) change in temperature. Multiplying by 1.165 five times gives an increase of approximately 2. Thus, the reverse saturation current approximately doubles for every 5°C temperature increase. (Experimental is closer to 8°C .)

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DEPENDENCE OF THE LARGE SIGNAL MODEL ON TEMPERATURE

Temperature Modeling of the PN Junction – Continued

- PN Junctions (Forward biased – v_D constant):

$$i_D \cong I_s \exp\left(\frac{v_D}{V_t}\right)$$

- Differentiating this expression with respect to temperature and assuming that the diode voltage is a constant ($v_D = V_D$) gives

$$\frac{di_D}{dT} = \frac{i_D}{I_s} \frac{dI_s}{dT} - \frac{1}{T} \frac{V_D}{V_t} i_D$$

- The fractional temperature coefficient for i_D is

$$\frac{1}{i_D} \frac{di_D}{dT} = \frac{1}{I_s} \frac{dI_s}{dT} - \frac{V_D}{TV_t} = \frac{3}{T} + \left[\frac{V_{Go} - V_D}{TV_t} \right]$$

- If V_D is assumed to be 0.6 volts, then the fractional temperature coefficient is equal to $0.01 + (0.155 - 0.077) = 0.0879$. The forward diode current will approx. double for a 10°C .

- PN Junctions (Forward biased – i_D constant): $V_D = V_t \ln(I_D/I_s)$

- Differentiating with respect to temperature gives

$$\frac{dv_D}{dT} = \frac{v_D}{T} - V_t \left(\frac{1}{I_s} \frac{dI_s}{dT} \right) = \frac{v_D}{T} - \frac{3V_t}{T} - \frac{V_{Go}}{T} = - \left[\frac{V_{Go} - v_D}{T} \right] - \frac{3V_t}{T} \approx -2.3 \text{ mV}/^\circ\text{C} \text{ if } v_D = V_D = 0.6\text{V}$$

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DEPENDENCE OF THE LARGE SIGNAL MODEL ON TEMPERATURE

Resistor Dependence on Temperature

- Diffused Resistors:** The temperature dependence of resistors depends mostly on the doping level of diffused and implanted resistors. As the doping level or sheet resistance increases from $100\Omega/\square$ to $400\Omega/\square$, the temperature coefficient varies from about $+1000 \text{ ppm}/^\circ\text{C}$ to $+4000 \text{ ppm}/^\circ\text{C}$. Diffused and implanted resistors have good thermal conduction to the substrate or well.
- Polysilicon Resistors:** Typically has a sheet resistance of $20\Omega/\square$ to $80\Omega/\square$ and has poor thermal conduction because it is electrically isolated by oxide layers.
- Metal:** Metal is often used for resistors and has a positive temperature coefficient.
- Temperature Coefficients of Resistors:** (ppm – parts per million)
 - n-well = $4000 \text{ ppm}/^\circ\text{C}$
 - Polysilicon = $500\text{--}2000 \text{ ppm}/^\circ\text{C}$
 - Metal = $+3800 \text{ ppm}/^\circ\text{C}$ (aluminum)
 - Diffusion = $+1500 \text{ ppm}/^\circ\text{C}$
 - Ion implanted = $+400 \text{ ppm}/^\circ\text{C}$

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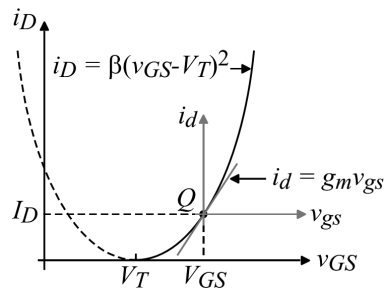
SMALL SIGNAL TRANSISTOR MODELS

What is a Small Signal Model?

- The small signal model is a linear approximation of a nonlinear model.
- Mathematically:

$$i_D = \frac{\beta}{2} (v_{GS} - V_T)^2 \xrightarrow{\text{Large Signal to Small Signal}} i_d = g_m v_{gs}$$

- Graphically



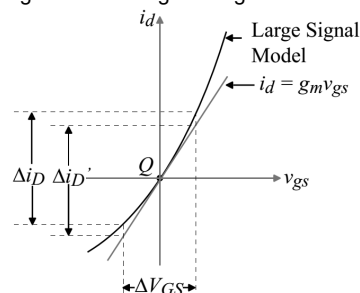
The large signal curve at point Q has been approximated with a small signal model going through the point Q and having a slope of g_m .

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SMALL SIGNAL TRANSISTOR MODELS

Why Small Signal Models?

- The small signal model is a linear approximation to the large signal behavior.
 - The transistor is biased at given DC operating point (Point Q above)
 - Voltage changes are made about the operating point.
 - Current changes result from the voltage changes.
- If the designer is interested in only the current changes and not the DC value, then the small signal model is a fast and simple way to find the current changes given the voltage changes.



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SMALL SIGNAL TRANSISTOR MODELS

How Good is the Small Signal Model?

It depends on how large are the changes and how nonlinear is the large signal model.

- The parameters of the small signal model will depend on the values of the large signal model.
- The model is a tradeoff in complexity versus accuracy (we will choose simplicity and give up accuracy).
- What does a simulator do? Exactly the same thing when it makes an ac analysis (i.e. frequency response)
- Regardless of the approximate nature of the small signal model, it is the primary model used to predict the signal performance of an analog circuit.

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SMALL SIGNAL TRANSISTOR MODELS

Small-Signal Model for the Saturation Region

- The small-signal model is a linearization of the large signal model about a quiescent or operating point.
- Consider the large-signal MOSFET in the saturation region ($v_{DS} \geq v_{GS} - V_T$):

$$i_D = \frac{W\mu_0 C_{ox}}{2L} (v_{GS} - V_T)^2 (1 + \lambda v_{DS})$$

- The small-signal model is the linear dependence of i_d on v_{gs} , v_{bs} , and v_{ds} . Written as,

$$i_d \approx g_m v_{gs} + g_{mbs} v_{bs} + g_{ds} v_{ds}$$

where

$$g_m \equiv \left. \frac{di_D}{dv_{GS}} \right|_Q = \beta (V_{GS} - V_T) = \sqrt{2\beta I_D}$$

$$g_{ds} \equiv \left. \frac{di_D}{dv_{DS}} \right|_Q = \frac{\lambda I_D}{1 + \lambda V_{DS}} \approx \lambda I_D$$

and

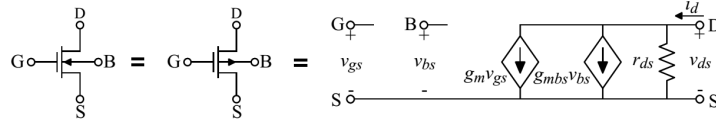
$$g_{mbs} \equiv \left. \frac{di_D}{dv_{BS}} \right|_Q = \left(\frac{di_D}{dv_{GS}} \right) \left(\frac{dv_{GS}}{dv_{BS}} \right) \bigg|_Q = \left(- \frac{di_D}{dV_T} \right) \left(\frac{dV_T}{dv_{BS}} \right) \bigg|_Q = \frac{g_m \gamma}{2\sqrt{2}|\phi_F| - V_{BS}} = \eta g_m$$

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SMALL SIGNAL TRANSISTOR MODELS

Small-Signal Model for the Saturation Region - Continued

- Complete schematic model:



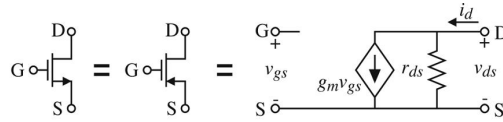
where

$$g_m = \left. \frac{di_D}{dv_{GS}} \right|_Q = \beta(V_{GS} - V_T) = \sqrt{2\beta I_D} \quad g_{ds} = \left. \frac{di_D}{dv_{DS}} \right|_Q = \frac{\lambda i_D}{1 + \lambda v_{DS}} \approx \lambda i_D$$

and

$$g_{mbs} = \left. \frac{i_D}{v_{BS}} \right|_Q = \left(-\frac{i_D}{v_{GS}} \right) \left(\frac{v_{GS}}{v_{BS}} \right) \bigg|_Q = \left(-\frac{i_D}{v_T} \right) \left(\frac{v_T}{v_{BS}} \right) \bigg|_Q = \frac{g_m \gamma}{2\sqrt{2|\phi_F| - V_{BS}}} = \eta g_m$$

- Simplified schematic model



- A very useful assumption: $g_m \approx 10g_{mbs} \approx 100g_{ds}$

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SMALL SIGNAL TRANSISTOR MODELS

Small-Signal Model for other Regions

- Active region: $g_m = \left. \frac{i_D}{v_{GS}} \right|_Q = \frac{K'WV_{DS}}{L} (1 + \lambda V_{DS}) \approx \left(\frac{K'W}{L} \right) V_{DS}$ $g_{mbs} = \left. \frac{i_D}{v_{BS}} \right|_Q = \frac{K'W\gamma V_{DS}}{2L\sqrt{2\phi_F - V_{BS}}}$

$$g_{ds} = \left. \frac{i_D}{v_{DS}} \right|_Q = \frac{K'W}{L} (V_{GS} - V_T - V_{DS})(1 + \lambda V_{DS}) + \frac{I_D \lambda}{1 + \lambda V_{DS}} \approx \frac{K'W}{L} (V_{GS} - V_T - V_{DS})$$

- Note: While the small-signal model analysis is independent of the amplitude of the signal, the small-signal parameters are not.

- Weak inversion region:

If $v_{DS} > 0$, then
$$i_D = I_t \frac{W}{L} \exp\left(\frac{v_{GS} - V_T}{nV_t}\right) \left(1 + \frac{v_{DS}}{V_A}\right)$$

Small-signal model:
$$g_m = \left. \frac{di_D}{dv_{GS}} \right|_Q = I_t \frac{W}{L} \frac{1}{nV_t} \exp\left(\frac{v_{GS} - V_T}{nV_t}\right) \left(1 + \frac{v_{DS}}{V_A}\right) = \frac{I_D}{nV_t} = \frac{qI_D}{nkT} = \frac{I_D}{V_t} \frac{C_{ox}}{C_{ox} + C_{js}}$$

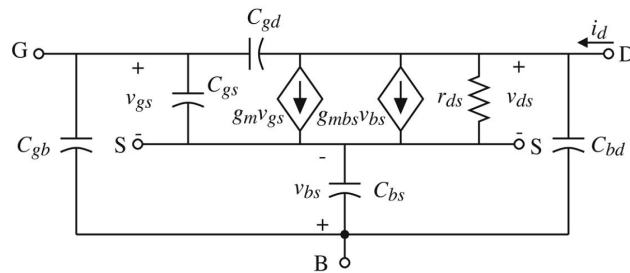
$$g_{ds} = \left. \frac{di_D}{dv_{DS}} \right|_Q \approx \frac{I_D}{V_A}$$

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SMALL SIGNAL TRANSISTOR MODELS

Small-Signal Frequency Dependent Model

- The depletion capacitors are found by evaluating the large signal capacitors at the DC operating point.
- The charge storage capacitors are constant for a specific region of operation.

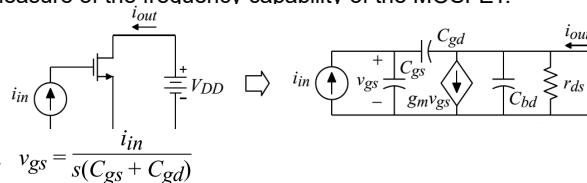


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SMALL SIGNAL TRANSISTOR MODELS

Gain-bandwidth of the MOSFET (f_T)

- The short-circuit current gain is measure of the frequency capability of the MOSFET.
- Small signal model:



- Small signal analysis gives,

$$i_{out} = g_m v_{gs} - s C_{gd} v_{gs} \quad \text{and} \quad v_{gs} = \frac{i_{in}}{s(C_{gs} + C_{gd})}$$

- Therefore,

$$\frac{i_{out}}{i_{in}} = \frac{g_m - s C_{gd}}{s(C_{gs} + C_{gd})} \approx \frac{g_m}{s(C_{gs} + C_{gd})}$$

- Assume $V_{SB} = 0$ and the MOSFET is in saturation,

$$f_T = \frac{1}{2\pi} \frac{g_m}{C_{gs} + C_{gd}} \approx \frac{1}{2\pi} \frac{g_m}{C_{gs}}$$

- Recalling that

$$C_{gs} \approx \frac{2}{3} C_{ox} W L \quad g_m = \mu_o C_{ox} \frac{W}{L} (V_{GS} - V_T) \rightarrow f_T = \frac{3}{4\pi} \frac{\mu_o}{L^2} (V_{GS} - V_T)$$

- For velocity saturation, $f_T \propto 1/L$.

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NOISE MODELS

Derivation of the MOSFET Thermal Noise Model

- In the active region, the channel resistance of the MOSFET is given from the simple large signal model as,

$$R_{channel} = \frac{1}{\left. \frac{\partial i_D}{\partial v_{DS}} \right|_Q} = \frac{1}{\frac{K'W}{L}(V_{GS} - V_T - V_{DS})} \approx \frac{1}{\frac{K'W}{L}(V_{GS} - V_T)} = \frac{1}{g_{m(sat)}}$$

The current thermal noise spectral density of a MOSFET in the active region would be

$$i_n^2(\text{active}) = \frac{4kT}{R_{channel}} = 4kTg_{m(sat)} \quad (\text{A}^2/\text{Hz})$$

- In the saturation region, approximate the channel resistance as 2/3 the value in the active region resulting in 2/3 the noise. Therefore, in saturation we have the current thermal noise spectral density as,

$$i_n^2(\text{sat}) = \frac{2}{3} i_n^2(\text{active}) = \frac{8kTg_{m(sat)}}{3} \quad (\text{A}^2/\text{Hz})$$

- Translating this drain current noise to the gate voltage noise by dividing by g_m^2 gives

$$e_n^2 = \frac{8kT}{3g_m} \quad (\text{V}^2/\text{Hz})$$

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NOISE MODELS

Derivation of the MOSFET Thermal Noise Model - Continued

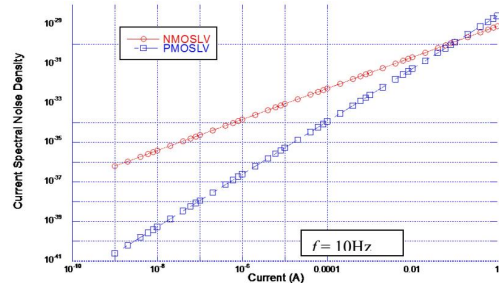
- The influence of the back gate on the thermal noise can be developed by replacing g_m of the previous expressions with $g_m + g_{mbs}$, we have the current thermal noise spectral density in saturation as

$$i_n^2 = \frac{8kT(g_m + g_{mbs})}{3} \quad (\text{A}^2/\text{Hz}) = \frac{8kTg_m(1 + \eta)}{3} \quad (\text{A}^2/\text{Hz}) \quad \eta = \frac{g_{mbs}}{g_m}$$

- 1/f Noise model: Another significant noise contribution to MOSFETs is a noise that is typically inversely proportional to frequency called the 1/f noise. This 1/f noise spectral density is given as

$$i_n^2 = \left[\frac{KF I_D^{AF}}{f^S C_{ox} L^2} \right]$$

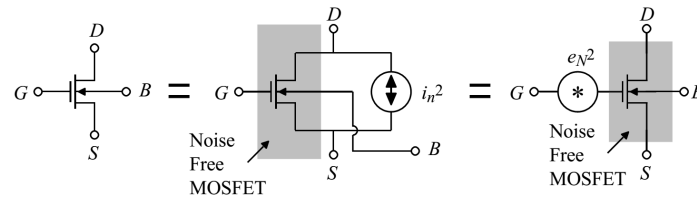
- KF = Flicker noise coefficient
- S = Slope factor of the 1/f noise
- Note:** The value of the 1/f noise for a PMOS is typically less than that for an NMOS for the same current and W/L.



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NOISE MODELS

MOS Device Noise at Low Frequencies



- The total current thermal noise spectral density in saturation is the sum of density of the back gate and 1/f noise model

$$i_n^2 = \left[\frac{8kTg_m(1+\eta)}{3} + \frac{KF I_D^{AF}}{\beta^S C_{ox} L^2} \right] \text{ (amperes}^2\text{/Hz)} \quad \eta = \frac{g_{mbx}}{g_m}$$

where

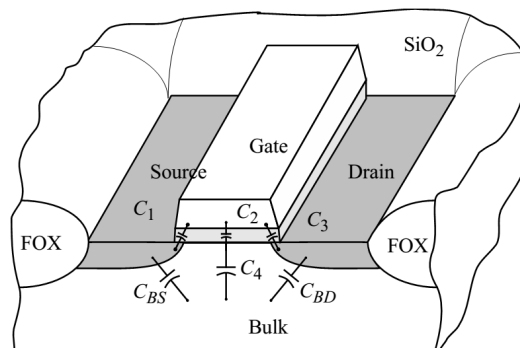
- k = Boltzmann's constant
- KF = Flicker noise coefficient
- S = Slope factor of the 1/f noise
- AF = Current coefficient

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MOSFET CAPACITOR MODEL

Submicron Technology

- Physical perspective:

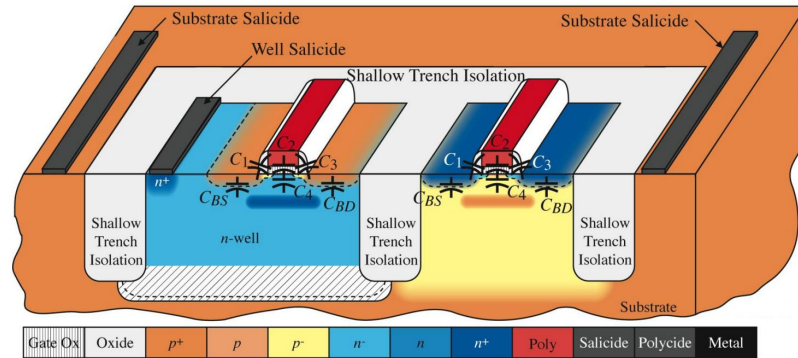


- MOSFET capacitors consist of:
 - Depletion capacitances
 - Charge storage or parallel plate capacitances

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MOSFET CAPACITOR MODEL

Deep Submicron Technology



- MOSFET capacitors consist of:
 - Depletion capacitances
 - Charge storage or parallel plate capacitances

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MOSFET CAPACITOR MODEL

MOSFET Depletion Capacitors

- Model

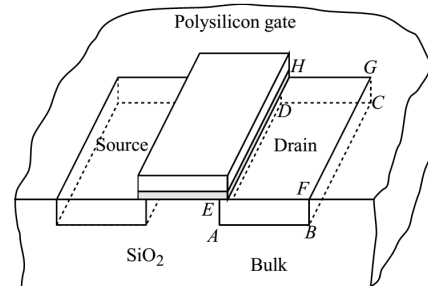
$$1. \quad v_{BS} \leq FC \cdot PB: \quad C_{BS} = \frac{CJ \cdot AS}{\left(1 - \frac{v_{BS}}{PB}\right)^{MJ}} + \frac{CJSW \cdot PS}{\left(1 - \frac{v_{BS}}{PB}\right)^{MJSW}}$$

$$2. \quad v_{BS} > FC \cdot PB: \quad C_{BS} = \frac{CJ \cdot AS}{(1 - FC)^{1+MJ}} \left(1 - (1+MJ)FC + MJ \frac{v_{BS}}{PB}\right) + \frac{CJSW \cdot PS}{(1 - FC)^{1+MJSW}} \left(1 - (1+MJSW)FC + MJSW \frac{v_{BS}}{PB}\right)$$

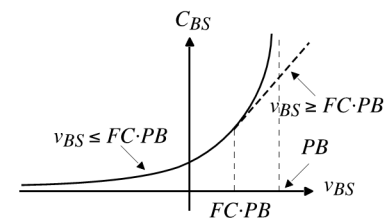
- where

- AS = area of the source
- PS = perimeter of the source
- CJSW = zero bias, bulk source sidewall capacitance
- MJSW = bulk-source sidewall grading coefficient

- For the bulk-drain depletion capacitance replace "S" by "D" in the above.



Drain bottom = ABCD
Drain sidewall = ABFE + BCGF + DCGH + ADHE



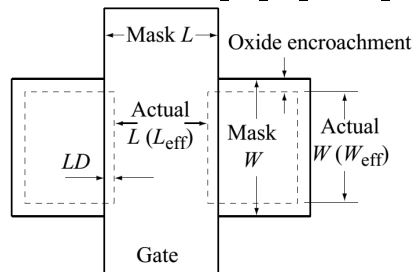
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MOSFET CAPACITOR MODEL

SM Charge Storage (Parallel Plate) MOSFET Capacitances - C_1 , C_2 , C_3 and C_4

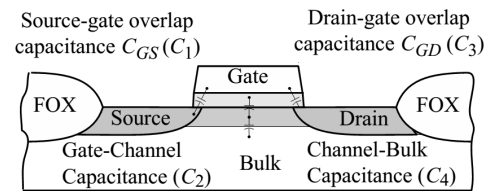
Overlap capacitances:

- $C_1 = C_3 = LD \cdot W_{\text{eff}} \cdot C_{\text{ox}} = \text{CGSO or CGDO}$
($LD \approx 0.015 \mu\text{m}$ for LDD structures)



Channel capacitances:

- $C_2 = \text{gate-to-channel} = C_{\text{ox}} \cdot W_{\text{eff}} \cdot (L - 2LD) = C_{\text{ox}} \cdot W_{\text{eff}} \cdot L_{\text{eff}}$
- $C_4 = \text{voltage dependent channel-bulk/substrate capacitances}$

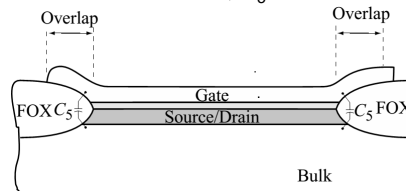


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MOSFET CAPACITOR MODEL

SM Charge Storage (Parallel Plate) MOSFET Capacitances - C_5

- View looking down the channel from source to drain, $C_5 = \text{CGBO}$



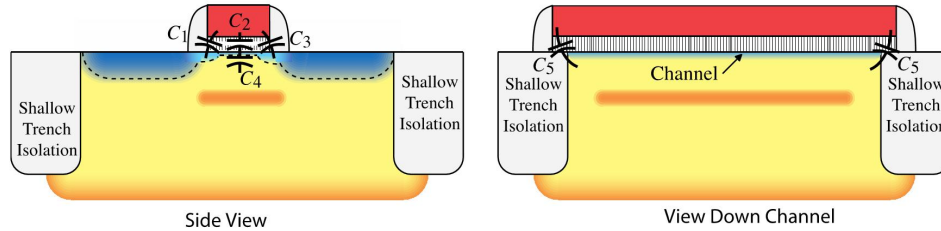
- Capacitance values based on an oxide thickness of $140\text{\AA}$ or $C_{\text{ox}} = 24.7 \times 10^{-4} \text{ F/m}^2$:

Type	P-Channel	N-Channel	Units
CGSO	220×10^{-12}	220×10^{-12}	F/m
CGDO	220×10^{-12}	220×10^{-12}	F/m
CGBO	700×10^{-12}	700×10^{-12}	F/m
CJ	560×10^{-6}	770×10^{-6}	F/m ²
CJSW	350×10^{-12}	380×10^{-12}	F/m
MJ	0.5	0.5	
MJSW	0.35	0.38	

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MOSFET CAPACITOR MODEL

DSM Charge Storage MOSFET Capacitances - C_1 , C_2 , C_3 , C_4 and C_5

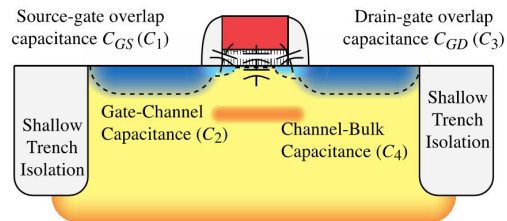
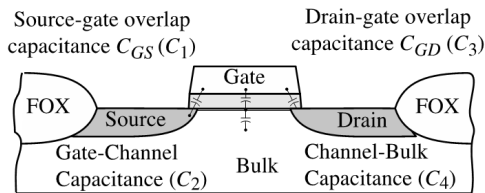


- C_1 and C_3 are overlap capacitors due to lateral diffusion of the source and drain
- C_2 is the gate to channel capacitance
- C_4 is the depletion capacitance between the channel and the bulk
- C_5 is the fringing capacitance between the gate and the bulk around the edges of the channel

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MOSFET CAPACITOR MODEL

MOSFET Capacitors for the Cutoff Region



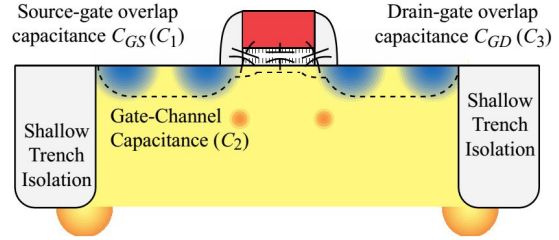
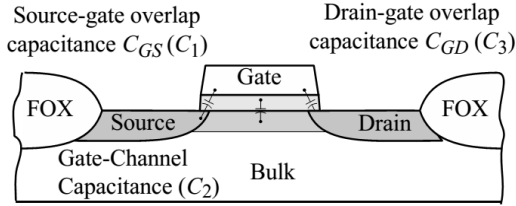
- As the gate-source voltage varies from 0 to V_T , the channel-bulk capacitor varies from a very large capacitor (because of a very small depletion region) to a capacitor much smaller than C_2 .
- Capacitors in Cutoff:

C_{GS}	$C_1 = C_{ox} \cdot LD \cdot W = CGSO \cdot W$
C_{GD}	$C_3 = C_{ox} \cdot LD \cdot W = CGDO \cdot W$
C_{GB}	C_2 varies from $C_{ox} \cdot L \cdot W$ to $2C_5$
C_{BD}	$C_{BD} = (CJ \cdot AD) / [1 - (v_{BD}/PB)]^{MJ} + (CJSW \cdot PD) / [1 - (v_{BD}/PB)]^{MJSW}$
C_{BS}	$C_{BS} = (CJ \cdot AS) / [1 - (v_{BS}/PB)]^{MJ} + (CJSW \cdot PS) / [1 - (v_{BS}/PB)]^{MJSW}$

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MOSFET CAPACITOR MODEL

MOSFET Capacitors for the Saturation Region



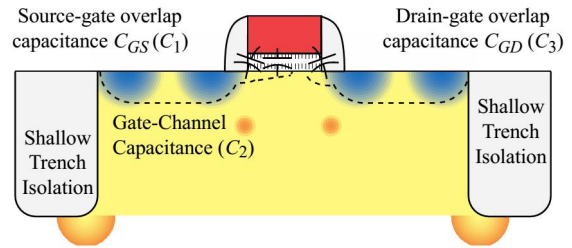
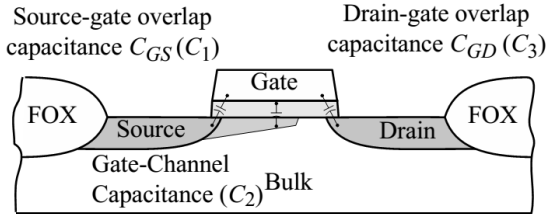
- In the saturation region, C_4 , becomes small and is not shown above.
- Capacitors in Saturation:

C_{GS}	$C_1 = C_{ox} \cdot LD \cdot W + (2/3)C_{ox} \cdot L \cdot W = [CGSO + (2/3)C_{ox} \cdot L]W$
C_{GD}	$C_3 = C_{ox} \cdot LD \cdot W = CGDO \cdot W$
C_{GB}	$2C_5 = 2 \cdot CGBO \cdot W$
C_{BD}	$C_{BD} = (CJ \cdot AD) / [1 - (v_{BD}/PB)]^{MJ} + (CJSW \cdot PD) / [1 - (v_{BD}/PB)]^{MJSW}$
C_{BS}	$C_{BS} = (CJ \cdot AS) / [1 - (v_{BS}/PB)]^{MJ} + (CJSW \cdot PS) / [1 - (v_{BS}/PB)]^{MJSW}$

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MOSFET CAPACITOR MODEL

MOSFET Capacitors for the Active Region



- In the saturation region, C_4 , becomes small and is not shown above.
- Capacitors in Saturation:

C_{GS}	$C_1 = C_{ox} \cdot LD \cdot W + (1/2)C_{ox} \cdot L \cdot W = [CGSO + (1/2)C_{ox} \cdot L]W$
C_{GD}	$C_3 = C_{ox} \cdot LD \cdot W + (1/2)C_{ox} \cdot L \cdot W = [CGDO + (1/2)C_{ox} \cdot L]W$
C_{GB}	$2C_5 = 2 \cdot CGBO \cdot W$
C_{BD}	$C_{BD} = (CJ \cdot AD) / [1 - (v_{BD}/PB)]^{MJ} + (CJSW \cdot PD) / [1 - (v_{BD}/PB)]^{MJSW}$
C_{BS}	$C_{BS} = (CJ \cdot AS) / [1 - (v_{BS}/PB)]^{MJ} + (CJSW \cdot PS) / [1 - (v_{BS}/PB)]^{MJSW}$

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**THE END
(THIS PART)**

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